Project Work

Information and Communication Systems

Chirp Modulation for Resilient Underwater Acoustic Communication

by

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Abstract

Underwater communication is difficult due to problems like multipath propagation, scattering, attenuation and others which forced to research for a resilient technique to communicate underwater efficiently and accurately enough so as to make surveillance top notch as the application space in shallow water is wide, ranging from pollution detection in dams, oil spills to monitoring shipping activity in ports and others. The micro acoustic underwater vehicles (μAUVs) are used in these applications because their low cost, small size and ready availability. Packet based preamble synchronization is a widely popular idea in order to synchronize two or more μAUVs prior to communication within them and this is used by research group smartPORT, Hamburg University of Technology. The modulation technique the institute is using at present to communicate is frequency shift keying (FSK) which is creating ambiguity in detecting the preamble boundaries due to multipath propagation and so chirp modulation is explored as the information in this case is spread over a frequency band rather than a single carrier frequency as is the case in FSK. The correlation results for a chirp also showed promise as it has a prominent peak in contrast to that of FSK, it is triangular shaped and due to multipath propagation, this triangle gets distorted and no prominent peak is found as in ideal case. So, chirp is explored in the project through different scenarios and experimental results are discussed to suggests that boundary detection efficiency and accuracy with chirp modulation is better rather than that of FSK in any given condition. Moreover, the effects of Doppler shift is investigated to check the robustness of chirp.

Table of Contents

List of Symbols

f_s	Starting frequency
f_e	End frequency
f_o	Observed frequency
f_t	Transmitted frequency
v_s	Speed of sound
v_t	Velocity of the transmitter
v_r	Velocity of the receiver
T	Pulse duration
P	Power
x	Message signal
c	Rate of change of frequency
R_{xx}	Cross-correlation of the message signal with itself

Chapter 1

Introduction

Underwater surveillance has ramped up in past few decades and is presently being applied in many different purposes. These manifold applications include localization of pollution in a dam, maintenance of oil rig situated off-shore which includes detection of oil-spill in the water, linking the submarines to on-shore control centers. In the port area, the application extends to automated tracking of ship movement, water quality, inspection of hull and others.

To inspect underwater aspects, autonomous devices should be employed readily and these specific devices are termed as μAUVs (Micro Autonomous Underwater Vehicles). In order to increase the lifetime of these devices underwater, the devices are low-powered. MONSUN, developed at University of Lübeck and Hippocampus constructed at Hamburg University of Technology are a few examples of μAUVs.

Communication between the μAUVs is an important to relay the information gathered during surveillance underwater. However, the underwater communication acoustic channel is severe, where the bandwidth is extremely limited and acoustic propagation better suited at lower frequencies. Adding to this, in case of shallow water communication, reflection due to the surface and thermoclines made the situation worse because of time-varying multipath propagation of acoustic signals.

Hence the primary idea is to communicate between the nodes (read as individual μAUVs) and the communication has to be resilient and robust. Packets are used to synchronize the nodes using preamble detection mechanism, which are in a mobile state underwater. Due to the effect described in earlier paragraph, wrong detection of symbols is a major problem with the state-of-the-art

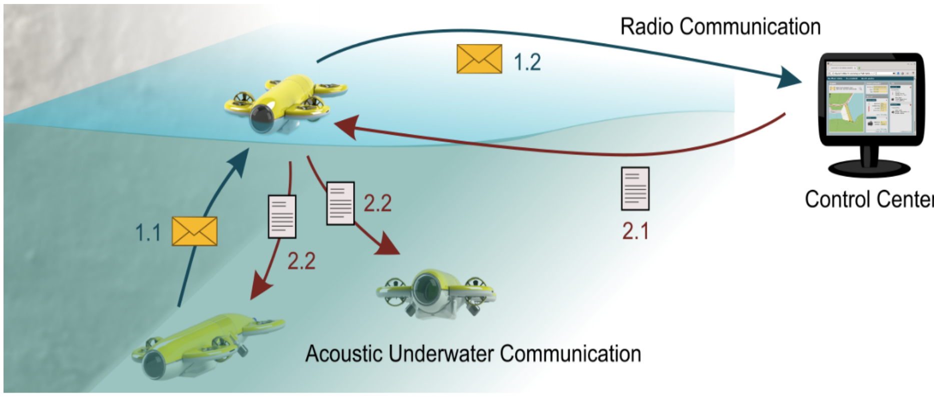


Figure 1.1: Swarms of MONSUN communicating underwater and relaying to surface hub for analysis.

technique (Frequency Shift Keying) used at Institute of smartPORT which is why the concept of chirp modulation, where the information is spread over a frequency band, is explored. This idea is motivated by the fact that whales and dolphin use the same for their communication underwater and in technological space, they are being implemented in radar applications since 1960s.

Henceforth, project is divided into four sections and will motivate the readers with background research, the experimental setup, test-case scenarios to real-time measurements, evaluation of received signals and concluding the work with contrasting features between chirp and frequency shift keying modulation.

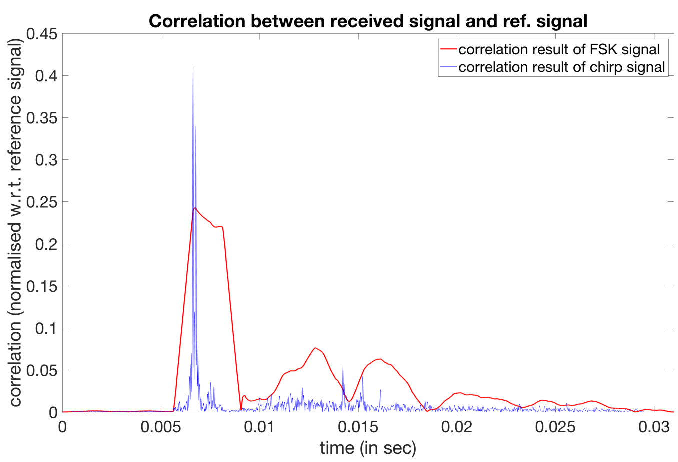


Figure 1.2: Correlation performed with both FSK and chirp received signals. Due to multipath propagation, the FSK symbol have slanted triangle which makes the detection ambiguous whereas for chirp received signals prominent peaks at time instant is found which helps in efficient detection and is observed to be multipath resistant as the line-of-sight peak is stronger that the reflected reception peak.

Chapter 2

Background and State-of-the-art

The discussion about the background is of paramount interest in developing into the research topic and its importance. Hence, this section will be divided into the need of the research project, the physical devices involved, the communication techniques needed and the mathematical basics required. Moreover, packet based synchronization will be briefly elaborated and the techniques required for detection of a received signal is also accounted later in the chapter.

2.1 Environmental Monitoring

Environmental monitoring is defined as systematic sampling of air, water, soil and biota in order to observe and study the environment as well as to derive knowledge from this process [2]. Environmental monitoring is multi-faceted and can be conducted on five different spheres of the Earth of which this project focuses on hydrosphere. Monitoring the water is a necessary component in order to analyze the localization of pollution, the marine life, tectonic plate movements and others. The state-of-the-art in case of shallow water environmental monitoring includes suspended particulate matter and oil detection [3]. To achieve this, underwater acoustics is proving to be an indispensable tool in surveillance. However, effective monitoring is challenging due to various environment constraints and thus requirement of rigorous scientific methods is a necessity for negotiating with the barriers poised by the surroundings.

The underwater environment has lot of variable such as ocean currents, attenuation, occurrence of multipath (refer Fig. 2.8) and difficulty in deployment of sensor nodes [4]. It is accordingly difficult and complex to gather and process the environmental information through underwater communication due to the aforementioned criterions unlike terrestrial wireless networks.

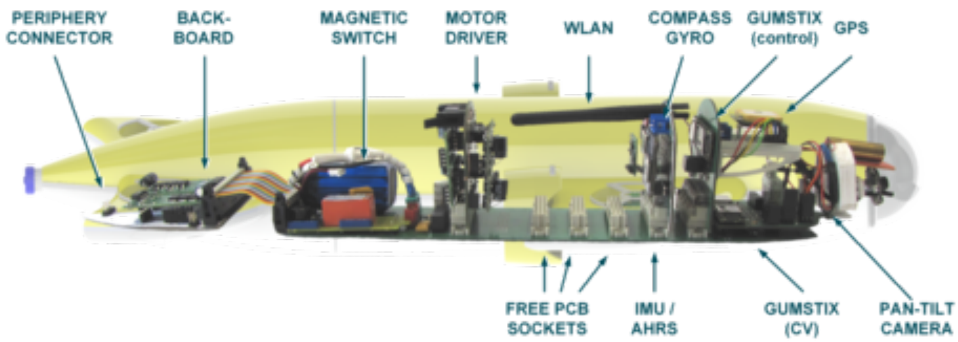


Figure 2.1: MONSUN with its housed components.

2.2 Micro Autonomous Underwater Vehicles

The sensor nodes which is mentioned in 2.1, are one of the important hardware involved in the process of environmental monitoring. These sensor nodes are housed in a waterproof shell along with the power source and communication module before being deployed in the water for measurements. Hence the overall device that is formed in the process is termed as micro autonomous underwater vehicles or μAUV in short.

The μAUVs gained a lot of attention lately from academics and research institutes which concentrates in underwater monitoring due to their small size, as the name suggests, robust architecture and unmanned nature of operation. In this research institute, MONSUN I & II and Hippocampus are two of such μAUVs used for inshore monitoring purposes. Figure 2.1 shows the MONSUN I μAUV in detail along with all its housed components. It has a length of 60 cm, 10 cm diameter at its widest cross-section and weighs 4kg and costs approximately €600. Four vertically mounted and two horizontally mounted motors are provided to allow the dynamic diving and rotations around the roll.

2.3 Acoustic Modem and Hydrophone

The modems used in case of acoustic measurements are slightly different from terrestrial communication and the institute of smartPORT have built an indigenous one for monitoring purposes.

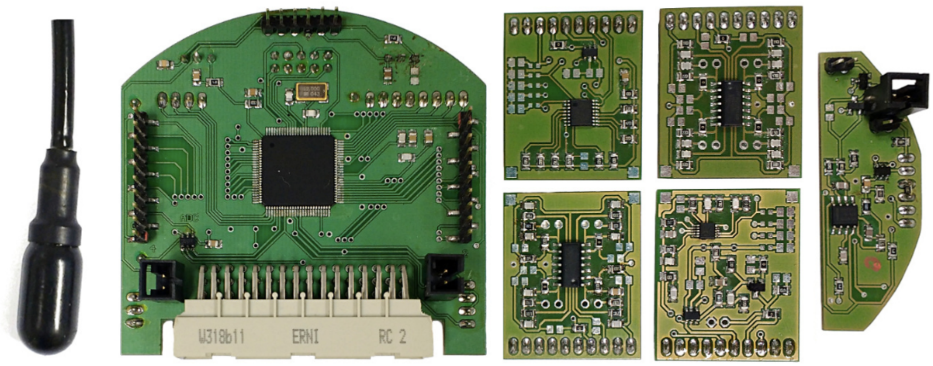
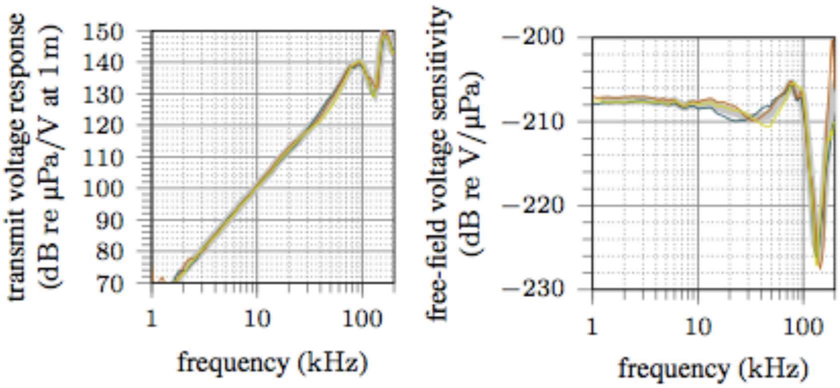


Figure 2.2: The smartPORT Acoustic modem prototype (left to right) – hydrophone, mainboard, amplifier (top-left), high-pass filter (top-right), low-pass filter (bottom-left), pre-amplifier (bottom-right) and power amplifier.

The microcontroller (μC) chosen for the modem was AVR32 with a maximum clock speed of 66MHz, a low-power DAC and ADC is used for to ensure high analog signal quality and low sampling noise and 12 GPIO pins are made accessible through pin headers to simplify debugging and evaluation. The above-mentioned are the core components and are supplied with two voltage converters - 3.3V for the μC and 5V for the analog receive circuitry. In addition to this, the sender module is attached with a power amplifier and receiver side with analog signal processing, where up to four analog filter stages can be connected to the main board. Figure 2.2 shows all the component used for the modem prototype and the assembled prototype has a size of 7 cm x 7 cm x 4cm [1].

In figure 2.2, on the leftmost side, a hydrophone is shown which is of type Aquarian Audio Miniature AS-1 [1] and this device act both as a transmitter and receiver. These hydrophones provide a bandwidth of 100 KHz with a flat-frequency response and a relatively high, linear transmit voltage sensitivity which is shown in figure 2.3.

The power amplifier is used to convert the DAC output to a maximum of 16V at 50mA at the sender module and can be disabled when it is not needed. The receive filter chain consists of four modules - pre-amplifier, high pass filter, low-pass filter and a post-amplifier. The pre-amplifier is electronically adjustable ranging from 40dB to 64dB with a step-size of 6dB whereas the post-amplifier ranges from 0dB to 38dB with a step-size of 4dB. Power amplifier and hydrophone allows a communication range of 100m [1].



(a) AS-1 as transmitter.

(b) AS-1 as receiver.

Figure 2.3: Transmit and receive characteristics of three Aquarian Audio AS-1 hydrophones.

2.4 Modulation Techniques

Communication between two μAUV nodes comes down to sending bit streams of 0s and 1s as required. To actually deliver the intended transmission, a carrier signal is needed which carries the information to be transmitted. The term modulation then can be defined as the process of variation of one or more properties of the carrier signal [6].

2.4.1 General Modulation Techniques

The most commonly used modulation is obtained by varying the amplitude, frequency or phase of the carrier signal and subsequently termed as amplitude, frequency or phase shift keying.

In Fig. 2.4, the concept of widely used modulation techniques are briefly introduced. Amplitude shift keying (ASK) is a technique where amplitude of the carrier is varied, i.e. when 1 is transmitted, there exists a modulated signal and for 0, nothing is sent out. However, for frequency shift keying (FSK), two different carriers are used for sending 1 and 0 and for phase shift keying (PSK), phase inversion occurs when the transmitted bit changes. For FSK, 1s are modulated with a carrier waveform with frequency of 50KHz and 0s are modulated with a carrier waveform of 30KHz.

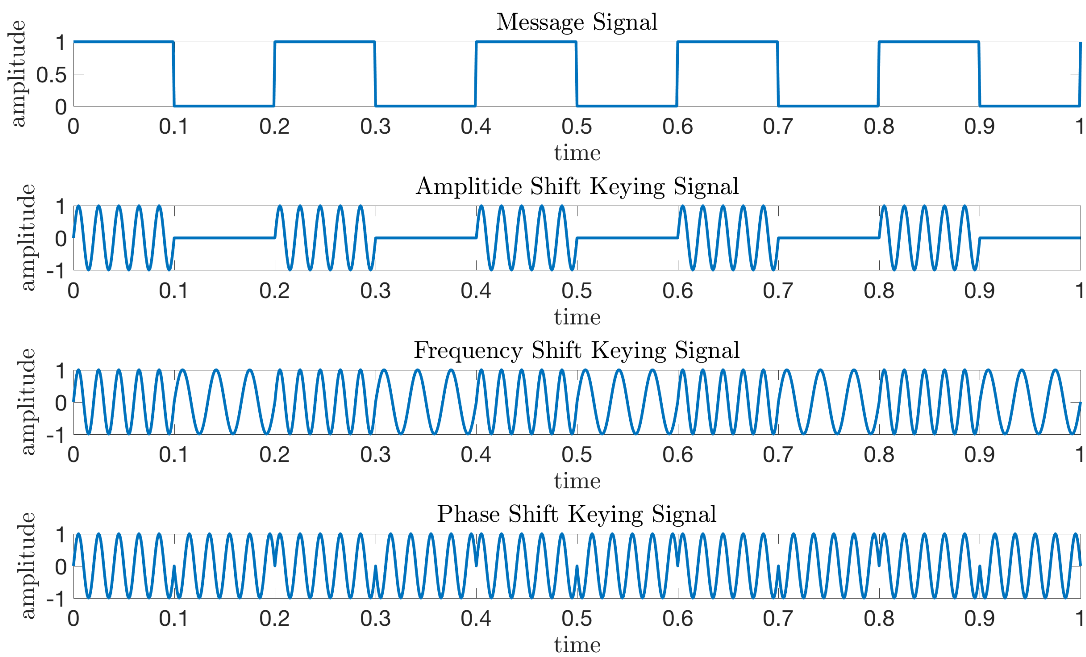


Figure 2.4: Modulated waveform generated by varying the amplitude, frequency, phase (top to bottom) of the carrier signal shown in Figure 2.4. All x-axes are in unit of time and y-axes are unit of volt.

ASK is not used in case of underwater acoustics because in this modulation nothing is sent for 0 and if the noise is too high, error in detection will occur. Also, due to the fact that a constant amplitude envelope in desired in wireless channel to place less strain on the power amplifier for transmission [7]. On the other hand, PSK inherently uses phase inverted carrier for modulating two different bits and if during transmission, there is a phase shift, again a detection error will result in. These observations with PSK and ASK left with FSK, which is used widely and is the state-of-the-art modulation technique used in the acoustic modem shown in figure 2.2.

2.4.2 Chirp Modulation

Chirp is another way to modulate the message signal which differs from the modulation techniques mentioned in section 2.4.1. While in FSK, PSK and ASK, the message signal is contained in a single frequency, it is spread over a frequency band in case of chirp modulation. The comparison

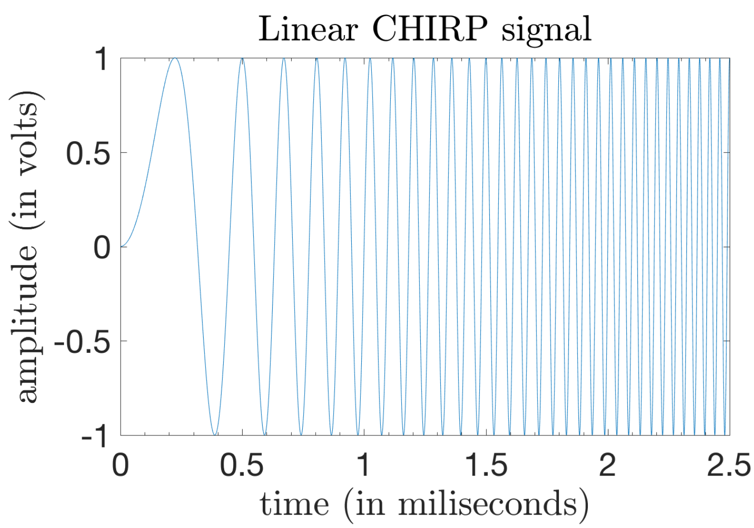


Figure 2.5: A linear up-chirp with sweep duration of 2.5 milliseconds and bandwidth 25KHz. The starting frequency is 0Hz and the ending frequency is 25KHz.

will follow this section and a short introduction of chirp modulation will be presented in this section.

CHIRP is the abbreviated form of Compressed High Intensity Radar Pulse, which is formulated by Sidney Darlington in 1954 and is used in RADAR signaling since World War 2 due to the requirement of long-range and high-resolution performance of the radar systems [8].

The mathematical definition of chirp is elaborated here. Let a waveform be defined as:

$$x(t) = \sin(\theta(t)) \tag{1}$$

So, the instantaneous frequency can be defined as:

$$f(t) = \frac{1}{2\pi}\frac{\mathrm{d}\theta(t)}{\mathrm{d}t} \text{ [9]} \tag{2}$$

and the frequency rate is formulated by:

$$c(t) = \frac{1}{2\pi}\frac{\mathrm{d}^2\theta(t)}{\mathrm{d}t^2} = \frac{\mathrm{d}f(t)}{\mathrm{d}t} \text{ [10]} \tag{3}$$

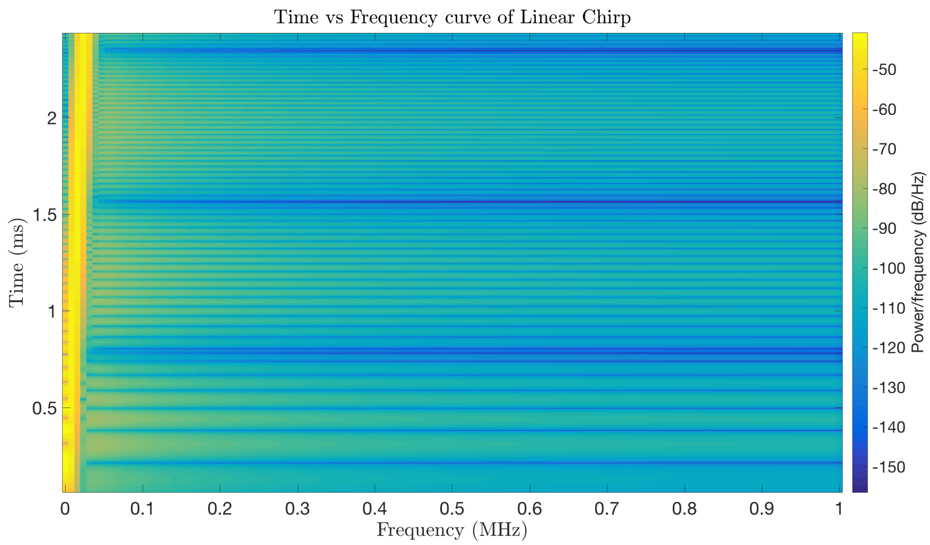


Figure 2.6: A linear up-chirp spectrogram obtained from Fig. 2.5. It shows the linear characteristics of time vs frequency of Linear chirp.

Chirp is classified on the basis of rate of change of frequency or the frequency rate in two types - linear chirp and exponential chirp. In case of linear chirp, the frequency rate changes linearly or can be seen as constant change of frequency and exponentially for exponential chirp. This change can be either increasing or decreasing and depending on it, chirp is divided into up-chirp and down-chirp [9].

For a linear chirp the rate of change of frequency can be denoted as:

$$c_l = \frac{f_e - f_s}{T} = \frac{BW}{T}, \tag{4}$$

and for exponential chirp, the same is given by:

$$c_e = \left(\frac{f_e}{f_s}\right)^{\frac{1}{T}}, \tag{5}$$

where f_e is the ending frequency and f_s is the starting frequency and T is, the time taken to sweep from f_s to f_e and BW is the bandwidth.

Hence, the corresponding time-domain function for the phase of linear chirp is:

$$\theta(t) = \theta_0 + 2\pi(f_s t + \frac{c_l}{2}t^2) \tag{6}$$

and that of exponential chirp is given by:

$$\theta(t) = \theta_0 + 2\pi f_s \left[\frac{c_e^t - 1}{\ln(c_e)}\right], \tag{7}$$

where θ_0 is the initial phase and the corresponding time domain function of a sinusoidal chirp can be denoted by:

$$x(t) = sin\left[\theta_0 + 2\pi\left(f_s t + \frac{c_l}{2}t^2\right)\right] \tag{8}$$

for linear chirp and by:

$$x(t) = sin\left[\theta_0 + 2\pi f_s\left(\frac{c_e^t - 1}{\ln(c_e)}\right)\right] \tag{9}$$

for exponential chirp. An example is shown in figure 2.6 for a linear chirp.

2.5 Acoustic Channel

Before delving into comparing between the state-of-the-art modulation technique and chirp, it is important to ponder upon the acoustic channel and some of its characteristics which are mentioned in chapter 1 and section 2.1.

2.5.1 Speed of sound

First, it is important to discuss on the speed of sound in water, which is the most important element for acoustic communication. Sound waves propagate with a compression and rarefaction through the water column, which is detected by hydrophone as a measure of pressure change. The speed of sound varies with temperature, pressure and salinity and below shows how speed of sound varies with the temperature [11].

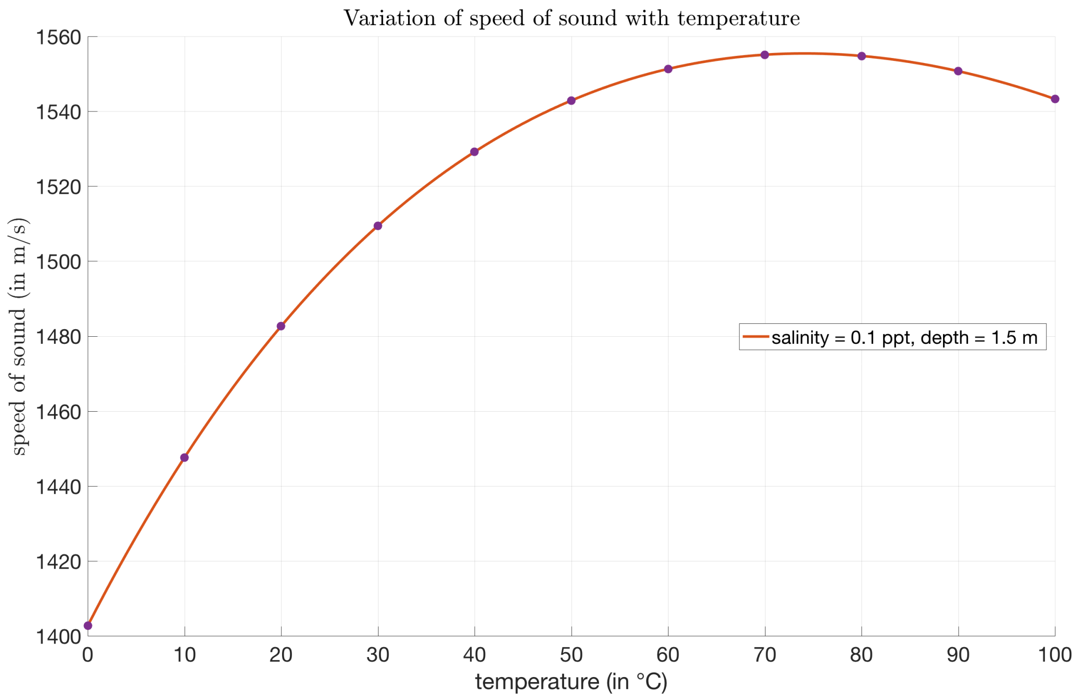


Figure 2.7: Change in speed of sound with respect to temperature keeping salinity and depth constant. Salinity considered here is for riverine channel surface as mentioned by Marine Research Council, India and the depth is taken to be 1.5m as per project limitation. The UNESCO equation presented in [12] is considered for the generation of the graph

Salinity varies water bodies and as the experiments are conducted in fresh water conditions, the salinity considered to be an average of 0.5 ppt. Depth is taken to be 1.5m as average due to the length restriction of the AS-1 hydrophones (figure 2.2) used in the experimental setup (figure 3.1).

2.5.2 Boundary Interactions

From figure 2.2, boundary interactions at the water surface is observed and the consequences are multipath propagation, scattering and consequently loss of portion of the message signal. It has also been pointed out in [13] that frequency dependent attenuation and interference affects the underwater channel. From [1] it is known that a signal level difference of 40 dB is found between the range 1 m and 100 m due to fading, which give rise to the need of adjustable amplifier gain.

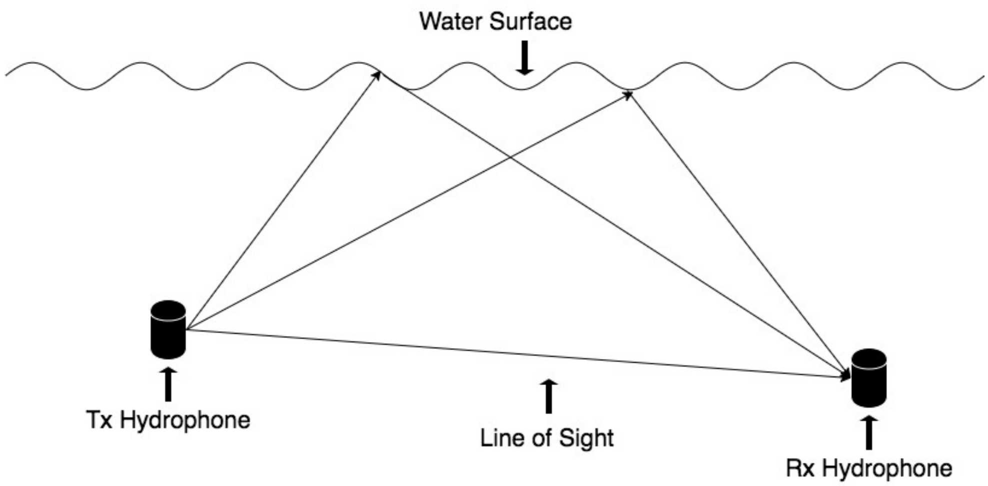


Figure 2.8: Surface reflection and multi-path propagations during transmission

Figure 2.2 also points out the multipath propagation and the echoes causes inter-symbol as well as intra-symbol interference at short distances, i.e. a few meters. This causes different reception at different point of time and so the resolution of the modulation used will affect in order to mitigate the effect of interference. The idea of resolution is further detailed in section 2.7.

The seabed interactions are not considered in this case because the project is limited to shallow water operation. Scattering of sound is another source of problem which is due to the biota and fish and hence the transient sounds result in decaying background which can be of much larger duration than the original signal.

2.5.3 Doppler Shift

For monitoring purposes, the μAUVs are in constant motion, which gives rise to a physical phenomenon known as Doppler shift. This essentially means that the received sound is somewhat different form the radiated or reflected sound. The relationship between the observed frequencies (f_o) and the emitted frequencies (f_t) when the receiving μAUV is moving at a velocity of v_r and transmitting μAUV is moving at a velocity v_t and the speed of sound considered to be v_s for that specific environment is given by [13]:

$$f_o = \left(\frac{v_s + v_r}{v_s + v_t}\right) f_t \tag{10}$$

The shift can be easily observed in a narrowband system because the transmitter frequency is known to the receiver and relative motion can be calculated [13].

If the speeds v_t and v_s are small compared to v_s, equation 10 can be reduced into:

$$f_o = \left(1 + \frac{\Delta v}{v_s}\right) f_t, \tag{11}$$

where $\Delta v = v_r - v_t$, and the change in frequency is calculated as:

$$\Delta f = \frac{\Delta v}{v_s} f_t, \tag{12}$$

Where $\Delta f = f_o - f_t$ is the amount of change in frequency.

To site the state-of-the-art research result noted in [1], the movement of the μAUVs lead to a frequency shift of 200Hz due to Doppler effect. This shift is accounted for a signal of 75KHz and two μAUVs moving in opposite direction at 2m/s.

2.6 Signal Processing

The modulation techniques, channel characteristics and hardware are described in section 2.1-2.4. But before comparing the FSK and chirp modulation, it is important to introduce the signal processing terminologies for a differentiation between the two.

The general components used for sound signal processing are shown in relation to section 2.3. The module of waveform generator helps in generation of the transmitted signal which is then power amplified (actual hardware shown in figure 2.2) and is put onto the channel by the hydrophones. From section 2.3, it is clear that the four modules: pre-amplifier, high and low pass filter and the post-amplifier acts as the signal conditioning block and the signal received is then available for processed to figure out whether the intended signal is received.

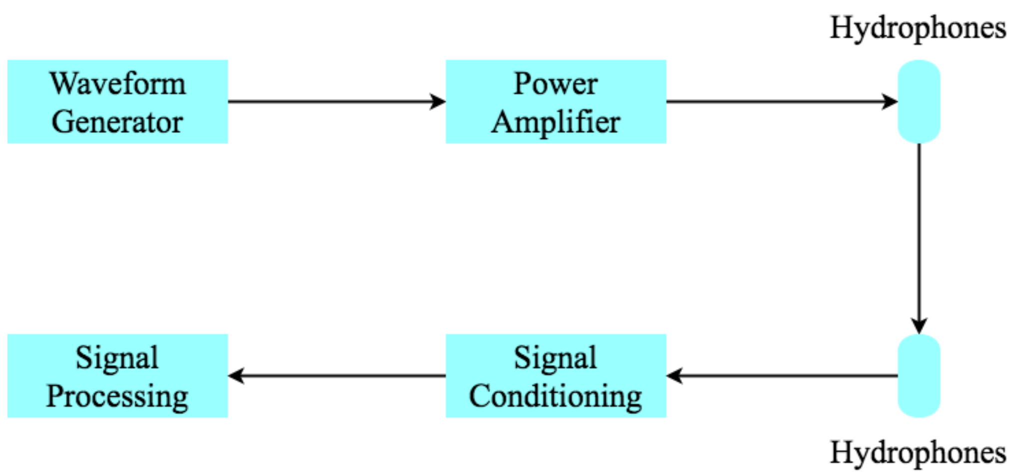


Figure 2.9: Components usually employed for sound transmission [11].

As the state-of-the-art technique is FSK, the time required for a complete cycle is of importance but the received signal in an acoustic environment consists of line-of-sight (LOS) reception with a delay along with the boundary interacted signal and lastly the decaying background as mentioned in section 2.5.2.

In order to figure out the intended reception of the transmission signal time-domain analysis and frequency-domain analysis of the received signal is explored. The idea of correlation is exploited to visualize the similarity between the radiated signal and the captured signal.

2.6.1 Correlation

Correlation may be seen as the measure of similarities between two given signals or statistics or more generally, two quantities. In the context of signal processing, correlation is classified into auto-correlation and cross-correlation. Autocorrelation can be formulated as [14]:

$$R_{11}(\tau) = \int_{-\infty}^{\infty} x_1(t)x_1(t-\tau)\mathrm{d}t \tag{13}$$

and cross-correlation can be seen as the measure of similarity of two different signal and mathematically defined by [14]:

$$R_{12}(\tau) = \int_{-\infty}^{\infty} x_1(t) x_2(t-\tau) \mathrm{d}t \tag{14}$$

Ideally, it is expected to get a peak response when autocorrelation is performed which will help in determining the transmitted signal at the receiver and the delay in reception. Hence this method motivates the distance between the receiver and the transmitter along with detection of the symbol.

2.6.2 Fast-Fourier Analysis

The frequency-domain representation of a signal carries information about signal's magnitude and phase response at each frequency. A time-domain signal can be converted into the frequency space with the help of a pair of mathematical transform: Fast Fourier Transform (FFT) and Inverse Fast Fourier Transform (IFFT). This analysis will also help to know the energy distribution of the signal over the frequencies [6].

The output of an FFT computation is complex as it involves knowledge at each frequency. Hence a signal can be defined as:

$$x = x_r + jx_i \tag{15}$$

and the magnitude is computed as:

$$|x| = \sqrt{(x_r^2 + x_i^2)} \tag{16}$$

and the phase is then obtained by:

$$\theta = \arctan \left({}^{x_i}/_{x_r} \right) \tag{17}$$

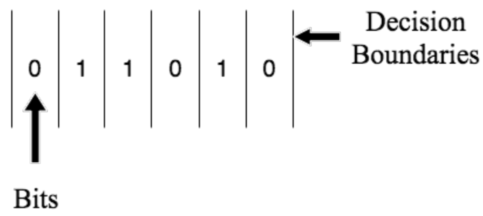


Figure 2.10: A probable Transmit order of bit and boundary separating two consecutive symbols. The preamble contains random sequence as shown and it is important to decide the boundary between two consecutive bits which ultimately help in distinguishing the preamble. This figure is explained in further detail in Fig. 3.3.

where x is a certain signal and x_r is the real part of the signal, x_i is the imaginary part and j is the complex number defined as $\sqrt{-1}$.

2.6.3 Preamble Detection and packet based synchronization

A significant amount of present day wireless communication systems is based on packetized communication and a correlation based detection is employed to detect the boundary of the preamble [15]. This technique is state-of-the-art communication architecture of the Institute of smartPORT. Hence the detection base on correlation is employed to locate the boundaries of the preamble. If the preamble is broken down into most basic 0s and 1s, it is important to locate the boundary between two consecutive symbols.

Packet based synchronization is another term introduced in this section as it is required to make the communicating μAUVs synchronized before exchanging relevant data between them. There are few correlations based approach available which will be described hereafter.

In figure 2.11a, the received signal is correlated with a delayed version of the self and the beginning of the frame can be estimate as it will be significantly higher when the received signal is pushed through the correlator. In 2.11b, an analog filter is placed before the signal is pushed into the correlator which is correlated with the stored reference signal [16]. In both the cases, peak search is important to detect the exact boundary separating the two consecutive symbols.

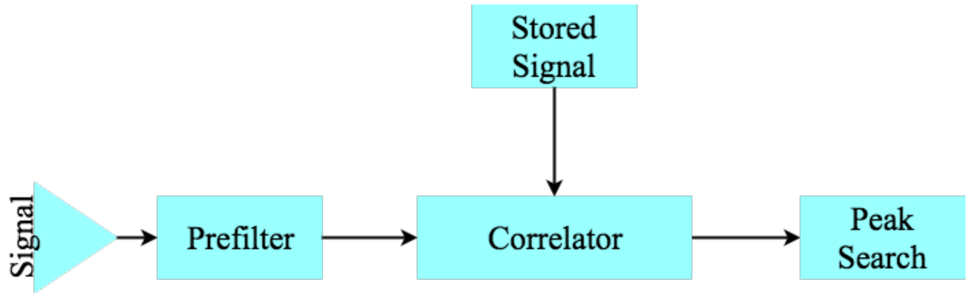


Figure 2.11: Frame detector with a pre-filter and a cross-correlator. The signal is passed through the pre-filter and then passed onto the correlator where it is correlated with the reference signal created with the a-priori information and search through to identify the peak.

Since the preamble is known a-priori, from [16] the problem with the 2.11a is found that the correlator will generate a triangular shaped output in case of an autocorrelation method and hence the reliability is poor which helped in developing method 2.10b and is more reliant with a cross-correlation based technique.

2.6.4 Sampling rate and step size of the stored signal at correlator

Sampling rate can be defined as samples considered per unit time in order to represent the signal digitally. The more the number of samples take per unit of time, the more accurate the signal will be. So, the choice of sampling rate is dependent on the hardware configuration as higher the sampling rate, more is the computation power required. The sampling rate of the stored signal can be reduced as an alternative approach by means of puncturing to find a trade-off between the computation time and accurate peak detection.

At correlator, the locally created signal is generated with a-priori information as described in section 2.6.3. This window of the correlator signal is another important consideration when computation power is concerned. The stored signal window can be shifted in larger step to decrease the computation time as multiplication in every step is not required.

Henceforth, sampling rate and step size is an important aspect to reduce the computation time of at the receiver without degrading the quality of peak detection.

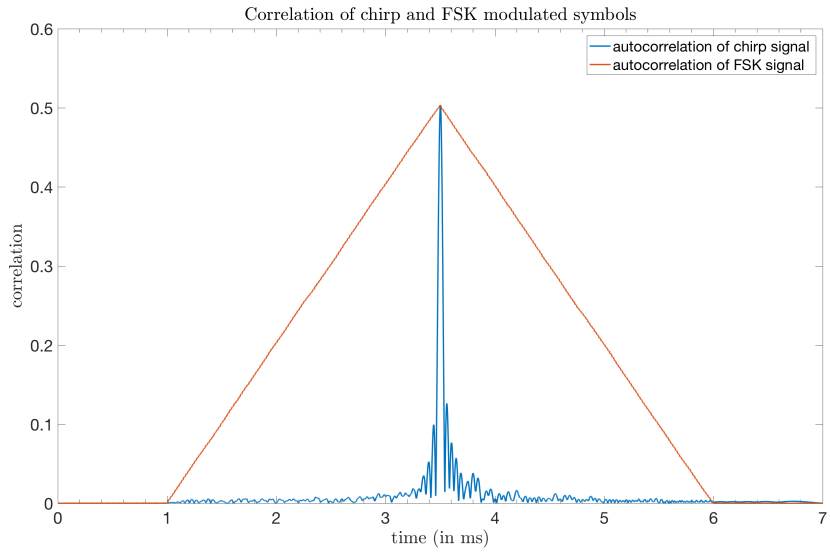


Figure 2.12: Comparison between the autocorrelation property of FSK and chirp signal. A symbol duration of 2.5ms is taken in consideration with carrier frequency for FSK is 25KHz and the bandwidth of chirp is 25 KHz sweeping from 0KHz to 25KHz.

2.7 Comparison between Chirp and FSK

To compare between Chirp and FSK, it is wiser to start with their individual autocorrelation characteristics, which follows the concept of CAZAC sequence, resolution and pulse compression. In case of FSK, the entire signal can be thought of as a rectangular window since only one frequency is present and it is known that correlation of two rectangular pulse is always a triangle. Now, when correlation is calculated according to equation 11, conforms with the theory of signal processing and is shown in figure 2.12.

The idea of packet-based synchronization as described in section 2.6.3, can be implemented only when the boundaries between the consecutive symbols are detected more or less accurately which is shown in figure 2.10.

For FSK, the demarcation of the decision boundary is ambiguous because of the triangular autocorrelation characteristics, and consequently, the detection probability of individual symbol will be erroneous. Moreover, the interference of the echo signal from the water surface cause the positive slope of correlation steeper or gentler based on the nature of the interference. A positive interference will result in steeper slope whereas a negative interference will provide a gentler slope. The problem is further motivated in section 3.1 with the boundary problem explained in detail.

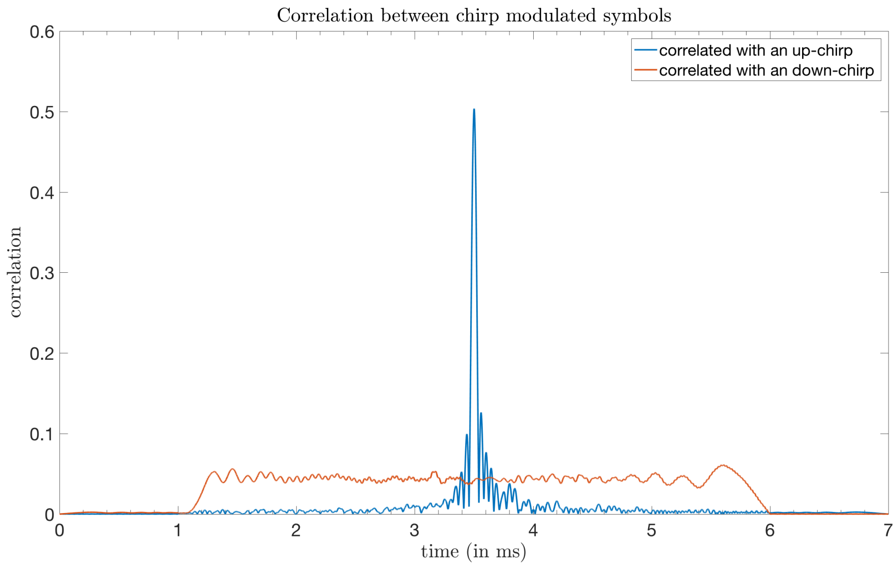


Figure 2.13: Correlation characteristics of Chirp. The time at which peak of the autocorrelation is observed is the time at which the whole signal is recovered. Cross-correlation is minimum ensuring optimal peak detection without ambiguity.

CAZAC sequence is the abbreviated form of Constant Amplitude Zero Auto-Correlation sequence. It is desired to have constant-amplitude over wireless channels as it places lesser strain on the power amplifier for transmission, which is why ASK is not used. The term zero auto-correlation states that for non-zero lag with out-of-phase periodic autocorrelation will amount to zero which leads to a simple detection structure of random access preambles as noted in [17]. Chirp is an ideal example of CAZAC sequence. Hence, from figure 2.12, it is observed that chirp has a strong prominent peak unlike FSK and this characteristic makes chirp pulse as an ideal cross-correlation as described in section 2.6.3 and shown in figure 2.11b.

The linear up-chirp signal as shown in figure 2.6 is also correlated with a linear down-chirp of same parameter set which helps in concluding the concept of CAZAC sequence as the peak is well distinguishable and is shown in figure 2.13. Hence, the preamble detection for packet based synchronization is ideally possible with the implementation of chirp and will yield a better result than that of FSK counterpart.

Next, is the time-bandwidth product. This is defined as the product of the duration of a signal and its spectral width. The time-bandwidth product for actual signals will vary but there will always be a minimum time-bandwidth product for the desired effect and while communicating over a channel, transmitting a certain amount of data over a given bandwidth requires a certain time.

The time-bandwidth product measures how well the available bandwidth for a given channel is used [18]. As for FSK, it is discussed in section 2.4.1 that only one type of carrier frequency is used to code one single bit and is shown in figure 2.4. However, in chirp modulation, the information is distributed over a frequency band and hence bandwidth plays an important role in this type of modulation. From the above paragraph, the time-bandwidth product can be exploited which is not available in case of FSK.

The time-bandwidth product of a chirp pulse is of genuine importance as it gives the compression ratio of the system and it approximately equates to the improvement of the SNR of the main lobe of the compressed with respect to relative to the original chirp.

The range resolution for an FSK with a sinusoidal pulse is:

$$\frac{1}{2}v_s T, \tag{18}$$

where v_s is the speed of sound wave and T is the pulse duration. Whereas, in case of chirp modulation with bandwidth Δf is defined as:

$$\frac{v_s}{2\Delta f} \tag{19}$$

The pulse compression ratio is formulated as:

$$\frac{T}{T'} = T\Delta f, \tag{20}$$

where $T' = \frac{1}{\Delta f}$ is the temporal width of the cardinal sine. The maximum of the autocorrelation function of the chirp signal is reached around 5ms in figure 2.12 and 2.13 and around this region the function behaves like a sinc function which is also known as cardinal sine [19].

The energy of the signal does not vary with signal compression; it just concentrates in the main lobe of the cardinal sine with an approximate width of T'. If P is the power of the signal before compression and P' be the power after compression and so the relation can be given by the following equation:

$$P' = P \times \frac{T}{T'} \tag{21}$$

The noise power does not change as it is uncorrelated against the transmitted pulse and hence after pulse compression, the power of the received signal can be considered being amplified by $T\Delta f$ [19].

2.7.1 Linear chirp and its Limiting factors

In this research context, linear chirp is chosen over exponential chirp. One of the reason is that the modulated waveform always coincides back in phase after the end of each period in case of linear change in frequency whereas for exponential change in frequency leads to change in its zero crossing [20].

Another reason and can be sited as a major advantage of using linear chirp is the Doppler Tolerance. In case of non-linear or exponential rate of change of frequency, even for lower values of Doppler, the main lobe (refer figure 2.12 or 2.13) will broaden resulting in raising side-lobes' level [21].

The limiting factors encountered with linear chirp is as follows and careful consideration of parameter choices are made based on these findings.

First of all, the Fresnel Ripples which causes in pulse degradation. The spectrum of linear chirp is rectangular only when the time-bandwidth product is large, i.e. when time x bandwidth exceeds 100. If the product is small, the rectangular profile will be degraded by the Fresnel ripples along with the matched filter [22].

Secondly, the side-lobes of the compressed chirp pulse can be minimized with different window function with the trade-off being broader main-lobe [21].

Thirdly, linear chirp is found very tolerant to Doppler shift with time-bandwidth product being less than 2000 [21].

Chapter 3

Motivation

The project is motivated due to two factors which is described in Fig 2.10 and Fig. 2.12. In 2.10, the boundary between two consecutive signal is of primary concern due to the correct detection of the preamble. This follows to the second argument which consolidates to the idea of correlation what helps us to find the peak. Here, the peak is to be understood as the point in time when the transmitted signal is completely received by the receiver. Hence, peak detection is a legitimate process by which the boundary of the preamble can be detected. In this project, the problem with FSK is that, certain portion of the signal can be lost due to the boundary interactions as mentioned in Sec. 2.5.2 and due to this the detection of the preamble boundary is not accurate and chirp is considered because it has a prominent peak and as shown in Fig. 2.12 and also due to the fact that up-chirp and down-chirp is quasi-orthogonal which means that if with the same a-priori information, a locally created down-chirp is correlated with the incoming up-chirp, a minimized correlation is observed and shown in figure 2.13 when compared with the correlation between locally created up-chirp with the received up-chirp.

So, before escalating to the setup of the project, the problem with FSK is further described in this section and the idea of how the project is proceeded is shown in detail and hence this chapter is divided into simulation results with FSK and chirp modulation and then how the whole experiment is setup considering the motivating plot, i.e. the idea of considering chirp over the state-of-the-art technique of FSK.

3.1 Boundary detection of the preamble sequence

The backdrop of the problem is introduced in Sec. 2.6.3 and by the diagram 2.10. From Fig. 3.1

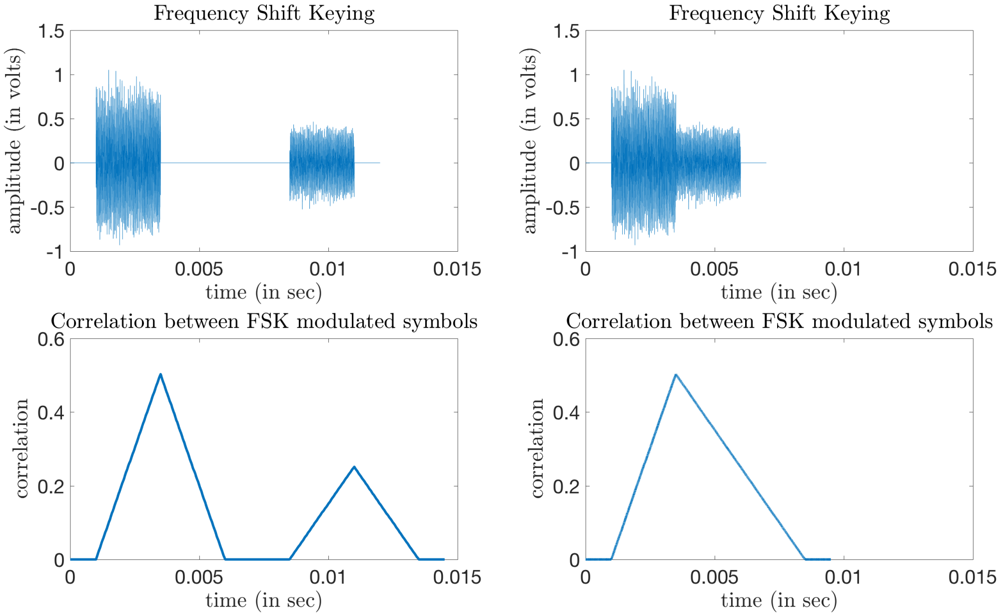


Figure 3.1: From left to right: two echo signals are separated enough and two prominent peak are found and on the right two echo signals are very close and hence they are mixed and so the negative slope of the triangle decrease slowly.

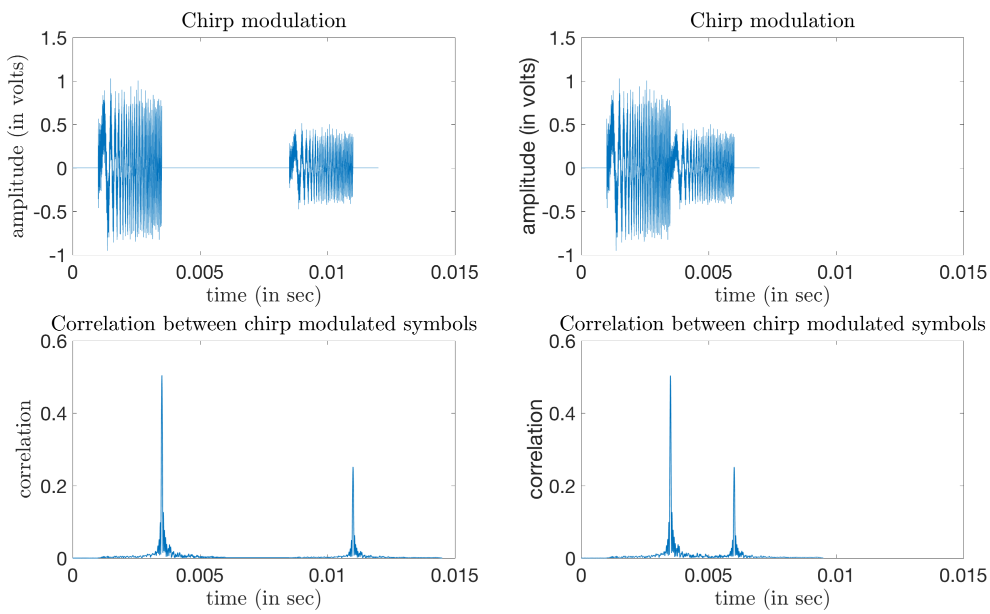


Figure 3.2: From left to right: two echo signals are separated enough and two prominent peak are found and on the right two echo signals are very close but still each of the signals are distinguished separately.

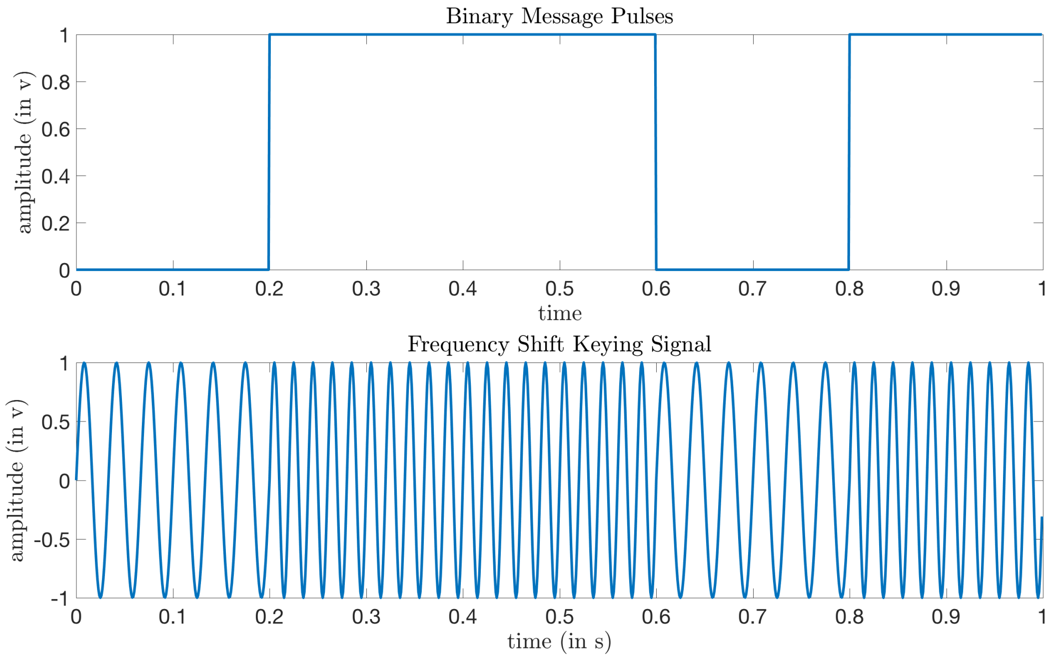


Figure 3.3: This figure is depicting the Fig. 2.10 in a signal space in an amplitude vs time plot and corresponding FSK modulation of each bit. The bit sequence as an example shown here is 01101 with each bit spanning a duration of 0.2 s.

and 3.2, it is observed that if the echo signals received due to the boundary interaction is separated in time, it is distinguished effectively for both FSK and chirp whereas when the echoes are overlapping or are very close to one another in time, the detection in case of FSK deteriorates rapidly rather when compared with chirp. In case of chirp, two prominent peaks are still found but not distinguishable for FSK.

This idea is taken forward to the boundary detection and reference is drawn from Fig. 2.10 which subsequently is shown in Fig. 3.3 in a more visual form. When the two ones are considered from the above Fig. 3.3, and then contrasted with the result that is obtained in Fig. 3.1 and 3.2, the problem is visualized immediately and the better expected performance enticed to continue with investigation of chirp for the project.

The better realization of prominent peaks in case of chirp even when two same signals are received almost one after the other may prove quite advantageous in detecting the boundary effectively than that of FSK modulation.

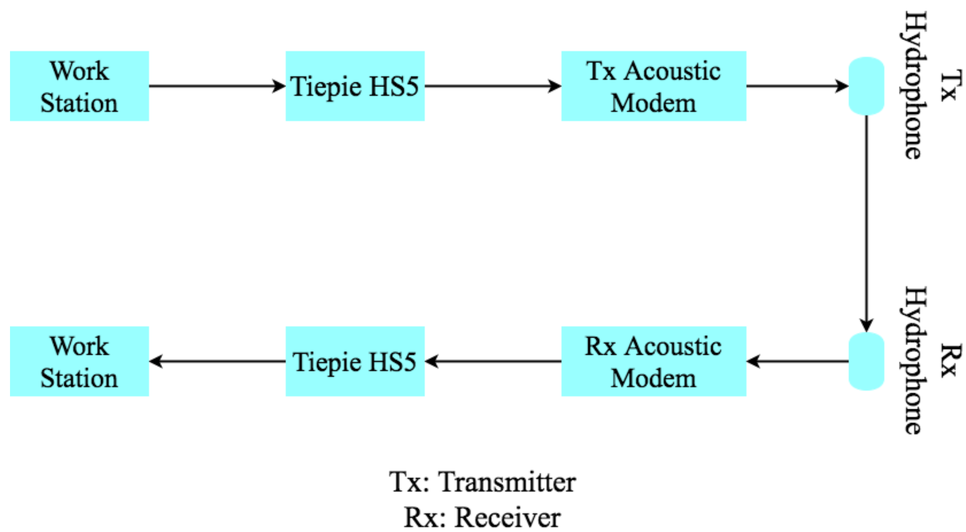


Figure 3.4: Overall setup for the experiment

3.2 Received signal analysis

In this project time-domain analysis of the received signal is done, i.e. cross-correlation with the received signal is carried out as shown in Fig. 3.1 and 3.2 and peak search is done to demarcate the preamble boundary. Moreover, the power spectral density of the received signal is observed to visualize the side lobe power in case of chirp. Test are done in various environments as mentioned in Sec. 3.3.6 and Doppler shift is also checked to introduce the robustness of chirp to the movements of micro autonomous underwater vehicles.

3.3 Setup of The Experiment

The setup of the experiment is similar to figure 2.9. The Tiepie Handyscope HS5 is used as a signal generator, which is enabled by a workstation and the power amplifier module mounted on the transmitting acoustic modem shown in figure 2.2 amplified the generated signal which is then sent to the hydrophone to ultimately put into the underwater acoustic channel. In the receiver front, the hydrophone picked up the signal as described in section 2.5.1 which is forwarded through the four-onboard module of the receiving acoustic modem acting as a signal conditioner and the conditioned signal is sent to the HS5 which consequently passed on to the workstation to get recorded and ultimately evaluated locally at the workstation.

The actual building blocks used for the entire experimental setup is shown in figure 3.4.

3.3.1 Parameters for Chirp modulation

The discussion in section 2.7 and subsection 2.7.1, paved the way for parameter selection. For the experiment, Linear chirp is selected over Exponential chirp as described in section 2.7.1. Afterwards, the selection of linear chirp motivates to choose the important parameter to contrast its selection with FSK modulation.

The first parameter considered is bandwidth, which is propelled by the definition of chirp as mentioned in chapter 1 and later prompted by equation 4. From the same equation, a dependence on the parameter time is noted which subsequently impelled in section 2.7, when the time-bandwidth product is defined. Based on that analysis, time or the duration of the signal is taken as the second parameter for the experiment. In figure 2.8 and chapter 1, the effect of reflection is talked about and afterwards the in section 2.5.2, boundary interaction is explained. This helped in considering parameters like silence before and after the chirp signal in order to check the fading time required for one chirp pulse. Moreover, to see the effect of multipath propagation as shown in figure 2.8, the experiment is iterated multiple times to observe the echo signals.

3.3.2 Tiepie Handyscope HS5

The idea of scrutinizing the HS5 is because of its difference with existing in-use HS3 from the same manufacturer. The highlighting difference between the two is the former being compiler independent, should one choose to use the HS5 with MATLAB. Even though both the device widely compatible in terms of programming platform integration, the approach of enabling and interacting with HS5 is completely different from HS3, which should be looked upon before moving from HS3 to HS5. The most important consideration with HS5 is to enable the generator before iterating through the variables rather than after setting the values to the variables else, the generator tends to forget the values and this results in recording only noise by the output.

The HS5 has three ports, of which one is output, demarcated as AWG, sends the arbitrary waveform generated (in this case a chirp pulse or a FSK pulse) to the acoustic modem. The other two ports acts as input ports and are designated as ch1 (channel 1) and ch2 (channel 2). Ch1 records the signal that is sent to the transmitter acoustic modem and ch2 records the signal from the filter/amplifier module of the receiver acoustic modem.

3.3.3 Gain of the filter module in Acoustic modem

In section 2.3, the gain levels available in the pre-amplifier and post-amplifier are mentioned. In this experimental setup, the gain stage of pre-amplifier is kept fixed and the post-amplifier is varied to examine the optimal reception of the signal, i.e. received signal should not be saturated in order to have the full signal at ch2 input (refer to section 3.2).

3.3.4 Choice of Values for the Parameter set

Chirp signals are generated with 14-bit ADC resolution and a maximum voltage range of ±8 volts and a sampling frequency of 2MHz. Signals are recorded with the same setup.

The duration of the chirp pulse is set at three different values - 1ms, 2ms and 5ms. The values are motivated by the state-of-the-art FSK symbol duration of 2.5ms. The other important factor to be observed is the effect of the time-bandwidth product. Next, the bandwidth of 25 KHz is chosen for the experiment with the frequency band 50KHz-75KHz. This bandwidth is then varied from 5KHz to 25KHz in order to analyze the time-bandwidth product.

The AVR32 microcontroller on the acoustic modem restricts the signal frequency to 75KHz, which is why state-of-the-art technique chose the above-mentioned band and in order to compare with the existing modulation, the bandwidth and the frequency band value is carried forward.

A silence period equal to 2.5ms is added prior to the signal to separate two consecutive pulse and a silence period of 25ms is padded after the signal to let the signal fade out completely. These silence period is to be observed in order to minimize the intersymbol and intra-symbol effects caused by the echoes from the water-surface interaction. The generated signal is amplified by 1.25v and a DC bias of 1.25v is added before sending it to the transmitting acoustic modem.

The whole experiment is repeated multiple times to visualize the difference in reception of the signal because of the reason mentioned in 2.5.2.

3.3.5 Algorithm for signal generation

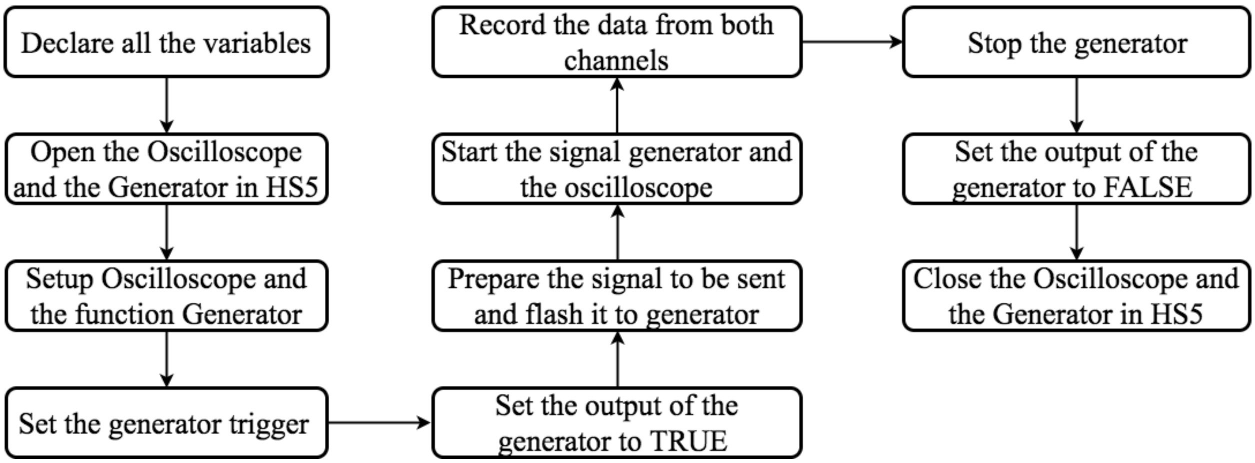


Figure 3.5: Flowchart for signal generation and recording the transmitted & received signal by Tiepie Handyscope HS5. The generator output should be turned on before preparation of signal but generator should be turned on only after flashing the signal to the generator.

3.3.6 Test Environments

For the validation of the project, tests are performed in real time environments like channel Harburg and Außenmühle Lake, Harburg. However different test environments are considered before conducting the outdoor tests.

The proper chirp signal generation and recording of the reception signal by HS5 are tested with a basic voltage divider circuit. Afterwards, the whole experiment is done in a small container, which exhibits the harshest environment in terms of reflection. Thereafter chirp test is carried out in a tank which is available in Hamburg University of Technology at Institute of Mechanical and Marine Technology.

Subsequently, when results were adequate, tests at channel Harburg and at the lake is conducted. In these scenarios, the chirp is compared with the FSK and recordings for both the modulation is taken with the same experimental setup.

Chapter 4

Test-case Scenarios

Examinations are conducted only with chirp in different test environment as mentioned in section 3.6. For this purpose, chirp signals are generated with 1ms of duration and with an available bandwidth of 25KHz where the starting frequency component f_s is 50KHz. Furthermore, this entire bandwidth of 25KHz it slotted into 1, 2, 3, 4, 5 segments keeping the starting frequency to be at 50KHz for each segment. Moreover, different duration of chirp signals, as mentioned in section 3.4, is also generated for efficient comparison.

Two test-case scenarios are explained in this section - small container test and tank test.

4.1 Small-Container Test

A small container was taken for the first set of our experiment. The dimension of the container was 19cm x _cm x _cm. The container is evidently small with respect to the real-time scenario and hence reflections will predominate while communication between the acoustic modems (shown in figure 10). The setup is exact as described in Fig. 3.1 and the test environment is depicted by Fig 4.1.

For the experiment, three different symbol length as mentioned in section 3.4 is considered with bandwidth 25 KHz and starting frequency 50KHz to compare the effect of time-bandwidth product in the following case. All other parameters are taken as mentioned in section 3.4.

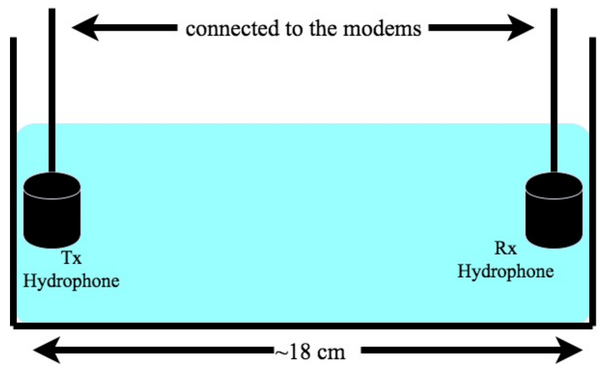
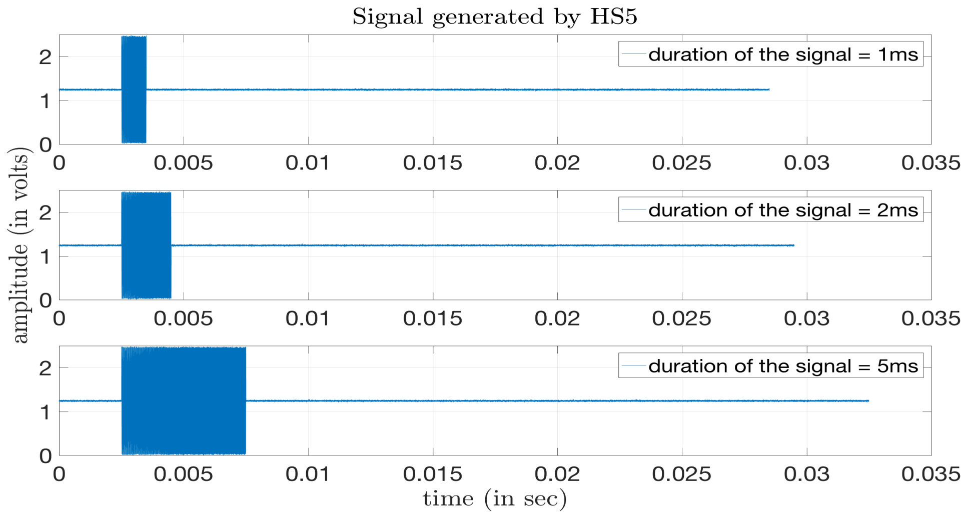


Figure 4.1: The hydrophones in the container are at approximately 18 cm apart from each other. The water is fresh tap water. The salinity is estimated at 0.1ppt and the depth of each hydrophone from the water surface is 2cm.

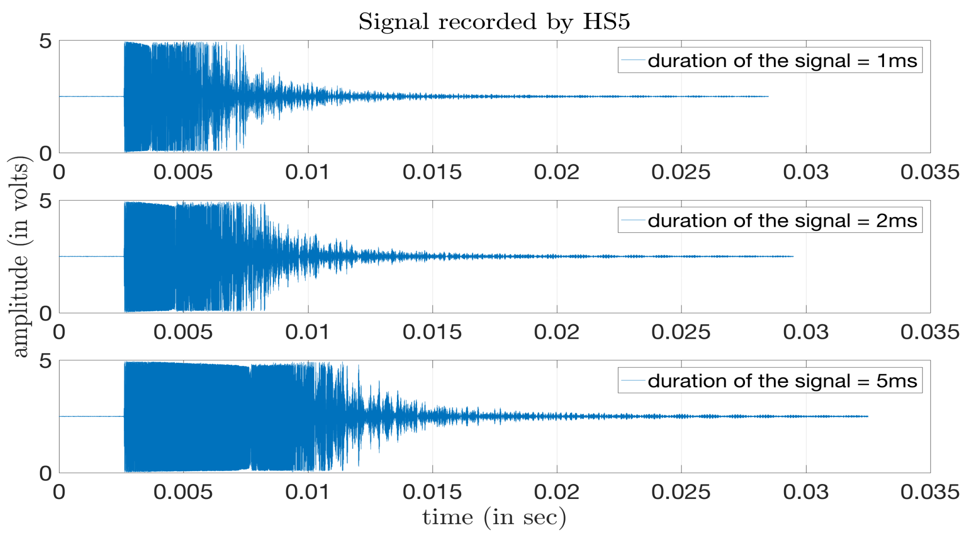
Line-of-Sight (LOS) communication was ensured in the small container with a small distance between the transmitter and the receiver. The surface on which the container was placed was bright gray and is mentioned in this case due to the fact that different profile of reflections in the received signal is observed with varied surface color. The container was on filled with tap water and except the impurities implicit particles present in the tap water, no other obstacles were put into the container.

The experiment is conducted with one iteration as no wave is generated manually in the container and only the robustness of decoding chirp was the main focus of observation. Though this environment is the worst among all, if detection of the transmit symbol is possible, the choice of using chirp can be justified.

Time-domain analysis is performed in this context and we apply correlation as mentioned in section 2.6.1 to detect the reception of the intended signal and as well as distance between the delay. The speed of sound for the discussed environment is found to be 1482m/s considering all the parameters, i.e. salinity of the water (0.01ppt), depth of the hydrophone (2cm), and the temperature of the room (20°C).



(a)



(b)

Figure 4.2: (a) The transmitted linear up-chirp signal with bandwidth 25 KHz, and starting frequency 50KHz. (b) The recorded received linear up-chirp signal with bandwidth 25 KHz, and starting frequency 50KHz.

An extremely short time delay time of 120μs is observed between the transmitted and received signal, is due to very short distance of 18 meters between the two hydrophones. The time-domain analysis as mentioned in section 2.6.1 is performed according to the method shown in Fig. 2.11b.

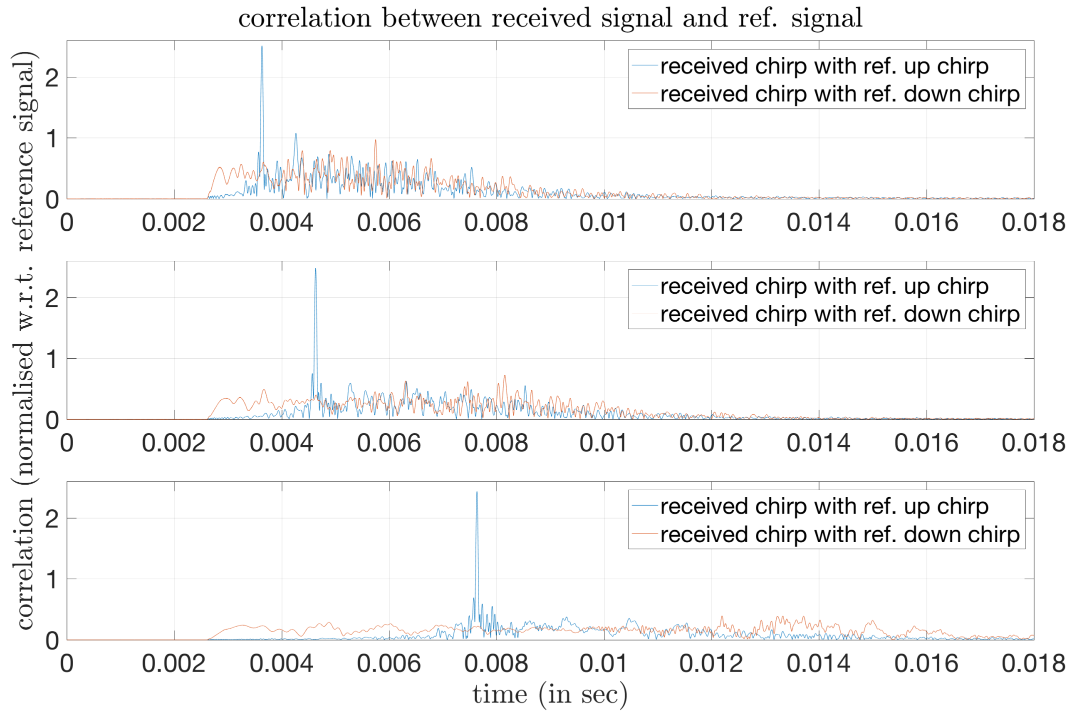


Figure 4.3: Correlation results when received data shown in figure 4.2b is correlated with the locally stored signal created with the a-priori information of the transmitted data at the receiver side. Duration of chirp pulse (Top to Bottom): 1ms, 2ms, 5ms.

Time-bandwidth product for each of the setup are 25, 50 and 125 respectively. From Fig. 4.3, it is observed that that the cross-correlation component, i.e. the correlation with the reference linear down chirp with the received linear up chirp is minimized in case of 5ms pulse duration than 2ms or 1ms pulse duration. The reason behind this is the longer the pulse duration, more is the power of the signal and so, 5ms pulse contribute to SNR more than other two pulses.

As described in section 2.7, the time-bandwidth product is proportional to the power concentration in the main-lobe and subsequently the side lobe will reduce. The power spectral density is shown below in Fig. 4.4. The purpose of considering the power spectral density function is to check the side lobe power. The Fig. 4.4 shows that the side lobe energy is lowered with increasing pulse duration.

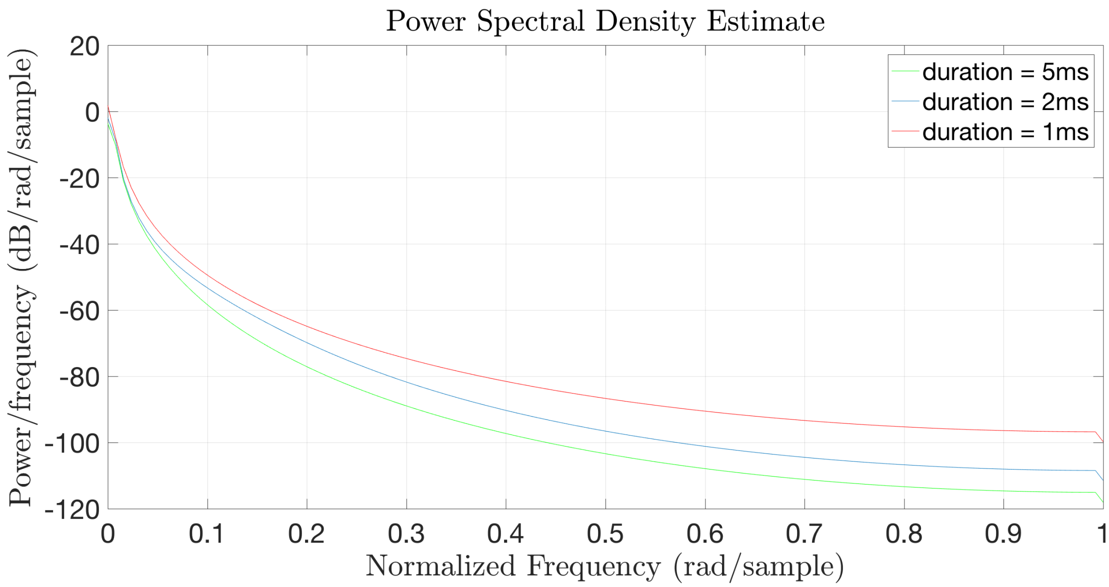


Figure 4.4: Power spectral density of the autocorrelation function shown in Fig. 4.3

The other thing that is observed from Fig. 4.2b is that the received signal seems a little saturated in the beginning which is why the gain of the post amplifier is varied to check the optimal gain to receive the whole signal. This is why when tank test is performed, the optimal gain is first checked so that the received signal is not saturated.

4.2 Tank Test

The second scenario considered was a tank in the Institute für Mechanik und Meerestechnik, TUHH. The water is particularly clear, with some instrument lying on the floor of the tank.

The gain level is varied to check whether the received signal is in saturation or not. From figure 18 it is noted that from a gain value of 38dB until 20dB, the line-of-sight signal is saturated and hence it is not wise to analyze those as the signal as a whole cannot be retrieved and ambiguity will be accompanied with the result. The received signal with gain value from 16dB to 0 dB seem viable for analysis and so the testing has been performed with gain values – 0dB, 12dB, 16dB and 38dB. The reflections are found out to be extremely strong as the receiver hydrophone was at the

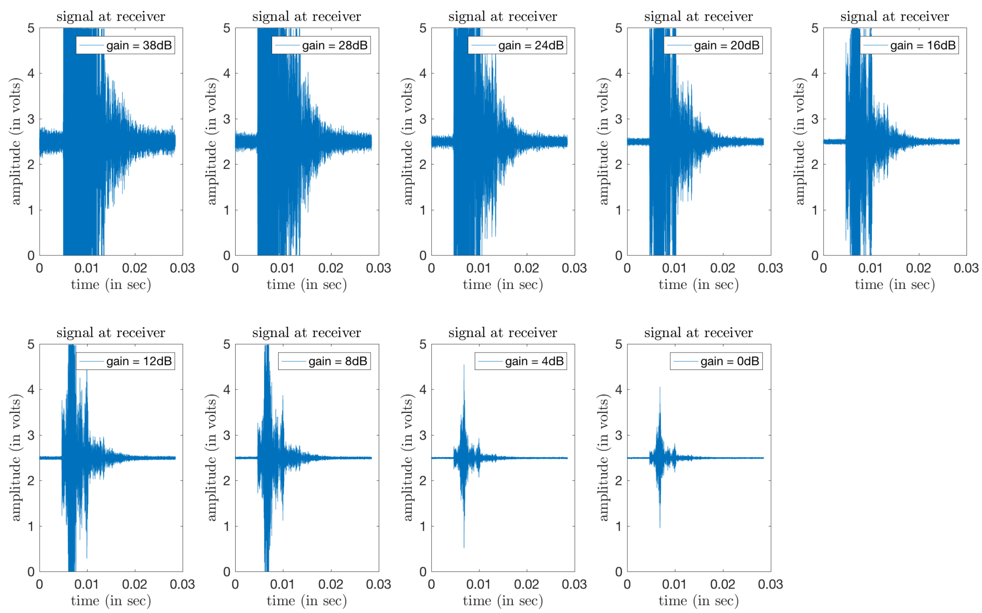


Figure 4.5: Received signal at different gain level of post-amplifier for a chirp signal of length 1ms and bandwidth 25KHz.

corner of the water tank and both the intersecting walls contributed to reflected received signals. In order to avoid such scenario because in open water it is unlikely to have such situation, the hydrophones are shifted away from corner positions and the experimental orientation can be seen in Fig. 4.5.

Due to the problem mentioned in earlier paragraph, the hydrophones are placed away from the walls to lower the effect of reflections. The position of both the hydrophones are shown in Fig. 4.6 and both of them were at a depth of 30cm.

In this scenario, chirp signals with 1ms duration are transmitted and the bandwidth are varied, starting with 25KHz and then reduced it down to 5 KHz. The intermediate bandwidth considered are 12.5KHz, 8.3KHz and 6.25KHz. Additionally, the distance between the hydrophones are larger compared to the small-container test and is shown in Fig. 4.6. Moreover, the waves on the surface instigated to iterate the experiment multiple times.

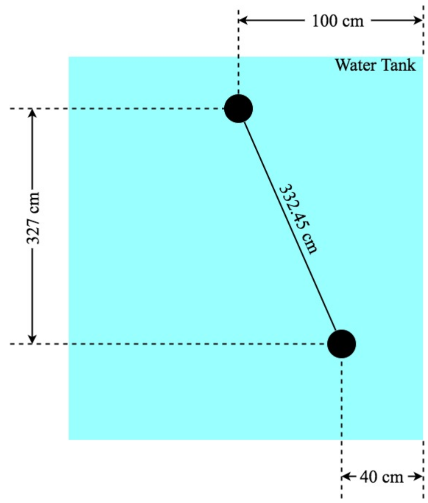


Figure 4.6: Top view of the water tank. Hydrophones are depicted as circles and they are kept away from the corners. A specific scenario is shown here and the experiment is done for two different distances between the two hydrophones.

From the Fig. 4.7, a delay of 2.21ms is observed in reception of the signal which almost conforms with the distance between the hydrophones, with an error in estimation is found out to be $2\mu s$. The correlation result is particularly important to visualize the exact time when the transmitted signal is received.

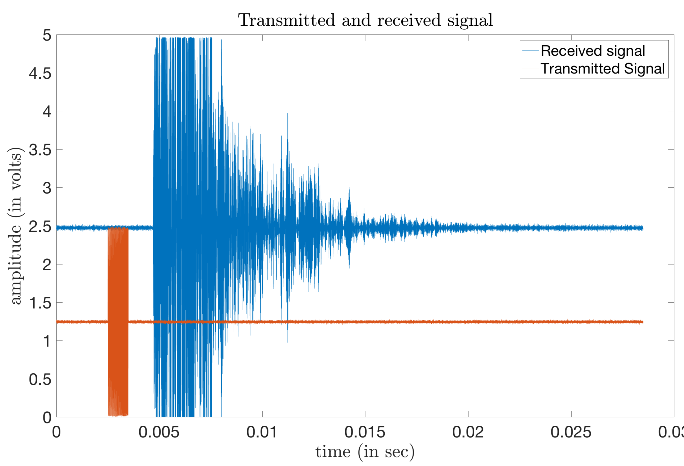


Figure 4.7: Transmitted and Received signal by the acoustic modems in the setup shown in Fig. 4.6. The chirp signal is of 1ms and bandwidth 25KHz with starting frequency 50KHz and with a post-amplifier gain of 16dB

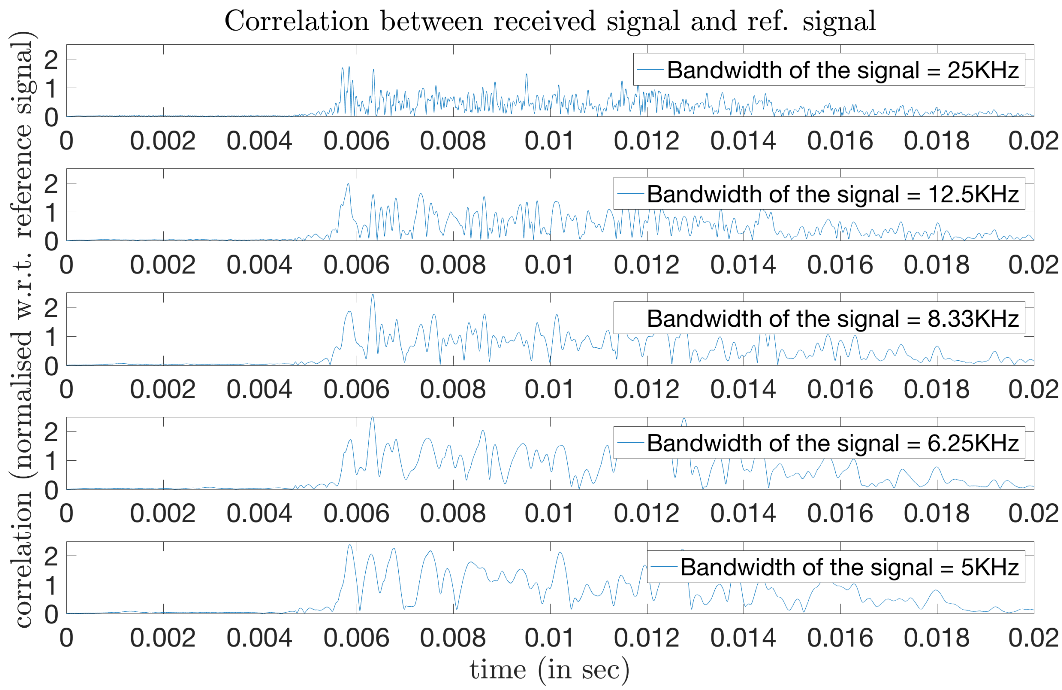


Figure 4.8: Correlation results when received data shown in tank setup is correlated with the locally stored signal created with the a-priori information of the transmitted data at the receiver side. Duration of Linear up-chirp pulse considered is 1ms and the ending frequency is 75KHz with post amplifier gain at 16dB. The first peak corresponds to the line-of-sight reception and a comparative study is shown later in Fig. 6.2 where the prominence to the LOS peak will be distinctly observed which is not in this case as the graph is flattened out for understanding the strength of reflection in a reflective environment.

The reference chirp is correlated with the received signal and as the delay is found from Fig. 4.7, it would be comfortable to decide if the received signal is detected accurately enough and if it is distinguished from the reflected signals. Hereafter, when the peaks are studied closely, the detection delay of 20μs is observed like the earlier case-study in small container test. Hence, a conclusion may be made on this delay are analog circuitry tolerance and the small error can be neglected.

From Fig. 4.8, it is observed that when the bandwidth is decreasing from top to bottom, the main lobe is increasing in width, i.e. the line-of-sight detection and side lobes are also increasing with time. This can be accounted due to decrease in time-bandwidth product from 25 down to 5. However, when the level of the first peak is considered for all the bandwidth, they all are above the value 1.7. Hence if a threshold level of the correlator is set at that value, detection can be accurate and computation will be faster as further correlation will not be necessary.

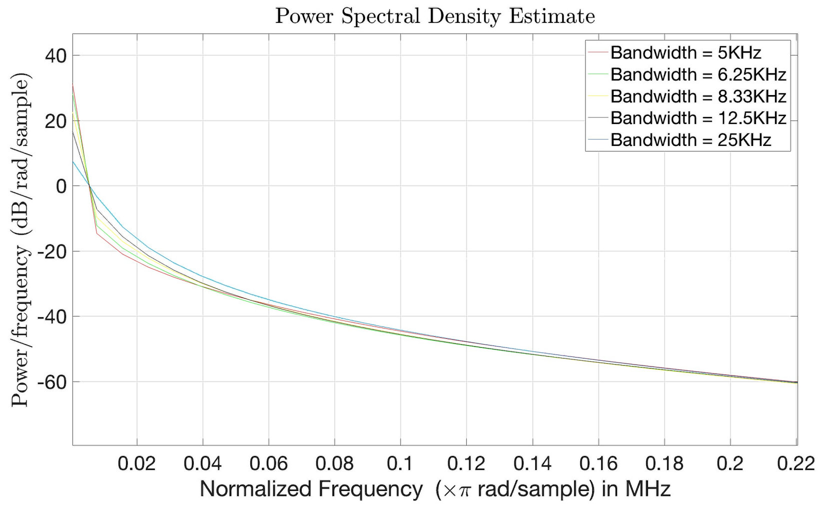


Figure 4.9: Power spectral density of the autocorrelation function shown in Fig. 4.8.

From the power spectral density (PSD) diagram shown in Fig. 4.8, it is observed that for a constant duration, with the decrease in bandwidth, the immediate side lobes' energy decreases rapidly which also ensures that for a specific duration of chirp pulse, bandwidth can be decreased without losing efficient detection.

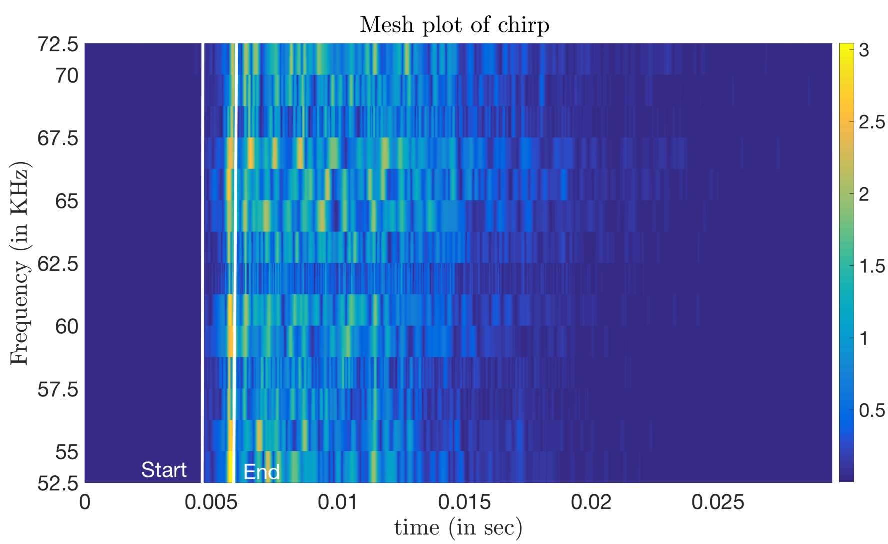


Figure 4.10: The received signal intensities (in terms of correlation values) of a chirp signal of 1ms in a distance of 3.3 m at 25 cm depth. "Start" is defined as when the signal reception started and "end", at which the maximum value of cross correlation occurs at the end of the received waveform, depicts the end of reception of the signal.

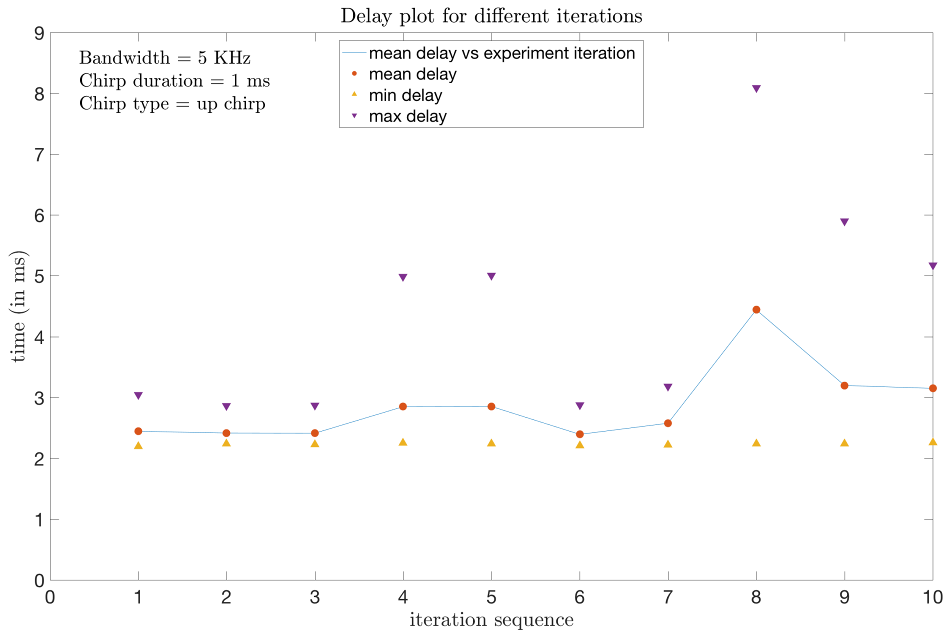


Figure 4.11: The average delay in estimating the reception of intended transmitted chirp signal of 1ms duration and a bandwidth of 5KHz, with starting frequency 50KHz. Each iteration has 5 transmitted linear up-chirp within the stipulated bandwidth of 25KHz (50KHz to 75KHz) and the maximum and minimum delay for each iteration are pointed out along with the average of each iteration.

In the Fig. 4.11, the delay of the smallest bandwidth is considered as from Fig. 4.10, it is seen that the side-lobe power in case of 5KHz bandwidth chirp diminishes almost equivalently as 25KHz chirp and so it might be wiser to use shorter bandwidth so that the small available bandwidth of 25KHz can be used more efficiently. Again, from the same figure, it is observed that minimum delay remains almost constant throughout all the iterations of the experiment done in the tank but due to the surface waves, maximum delay varies in each iteration. An overall average delay in detecting the transmitted chirp pulse of 2.9ms is found across all the iterations.

With all the data found in the test case provided an optimum motivation to test chirp in real world scenario base on three factors: negligible detection delay of $20\mu s$, which is described in Fig. 4.7 and Fig. 4.8, after taking the path delay into account, easy implementation of thresholding limits as the LOS component is fairly prominent in each scenario (refer Fig. 4.8) and even though the first reflection tend to overpower the LOS component with decreasing bandwidth, the first value crossing the threshold can be taken as the time at which the intended signal is detected. Lastly, a low overall delay in detection as depicted in Fig. 4.11 and effective side-lobe suppression as seen from Fig. 4.9 proved promising to continue with channel Harburg test.

Chapter 5

Real-World Test

The in-house test gave the anticipated results, which created enough confidence to test in real-world scenario. For this experiment, the channel of Harburg and Außenmühle lake in Harburg are chosen as it was convenient from Hamburg University of Technology.

5.1 Channel Harburg Test

The experiment was conducted on a boardwalk on the bank of the channel.

For this experiment, both FSK and chirp signals of 1ms duration are transmitted over the acoustic channel. The bandwidth of the chirp is varied as mention in Sec. 4.1 whereas for FSK, a frequency spacing of 400Hz is provided starting from 50KHz to 75KHz. All other parameters are consistent with Sec. 3.4.

The experimental setup was done with the hydrophone placed in such a way to have a line-of sight path for communication. The water is reed free, and the hydrophones are kept at a distance of 435cm and the transmitter hydrophone was at 100cm and receiver hydrophone was at 60cm. So, the effective distance between them was 461.73cm and taking the water temperature around 20°C, the speed of sound was taken to be 1482ms^{-1}. Hence the theoretical delay is 3.11ms.

When the received signal is zoomed in, the real-time delay is found out to be at 3.16ms and hence an error of 5μs in estimation is found which can be considered negligible.

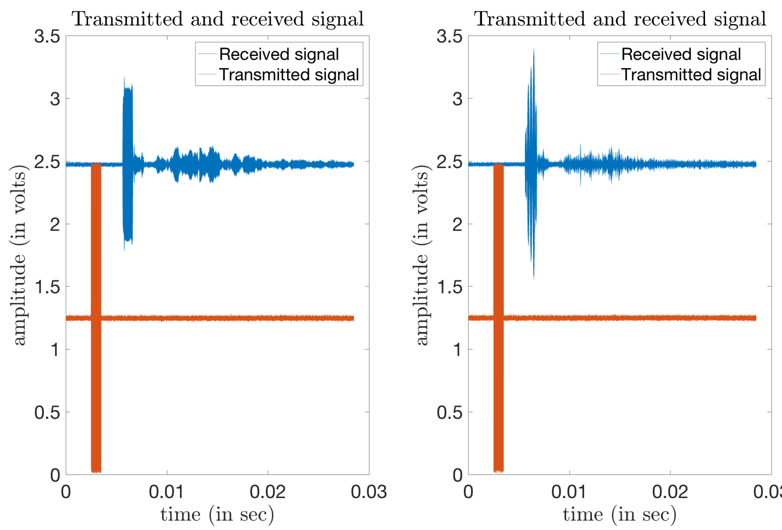


Figure 5.1: Transmitted and Received signals in channel Harburg. (Left to right) FSK signal is created with carrier frequency 75KHz and a chirp pulse of 1ms duration and bandwidth 25KHz with starting frequency 50KHz.

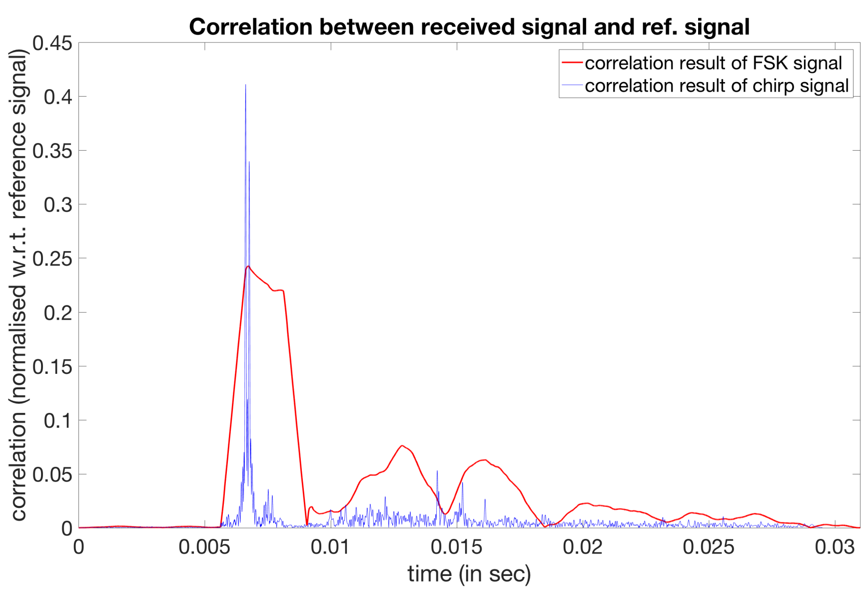


Figure 5.2: Correlation performed with both FSK and chirp received signals shown in Fig. 5.1. The correlation is only normalized with the locally generated data as the energy of the entire received signal is different in different point in time. FSK and chirp correlation for different

The time-domain analysis i.e. correlation is performed for both the signals to observe time instant of the peak, which is obtained at the point when entire LOS received signal is recorded at the receiver. The same is shown in Fig. 5.2 where it is visualized the kind of expected result as discussed in Sec. 2.7 and supported by figure 2.12. The peak in case of chirp signal is prominent whereas an area is obtained for FSK signal correlation. The discussion point for chirp would be the first reflection being comparable to LOS reception but in case of FSK, distinction between exact reception time is difficult as the reflected signal adds to the correlation, whose effect is also mentioned in [19].

Moreover, form Fig. 5.2 the threshold for detection should be placed at 0.35 for accurate detection of the received signal which is different as shown in figure 4.8, where the threshold detection was at 1.7 for efficient and exact detection. The argument for change in threshold level is that the threshold detection is to be defined as a function of post-amplifier gain. This is because the Fig. 4.8 is obtained at a gain of 16dB whereas in Fig. 5.2, the received signal is recorded for a post-amplifier gain of 0dB. The other is that the correlation is normalized only with reference to the locally created signal at the correlator as the received signal will be different in each recording because of the boundary interactions (Sec. 2.5.2).

The time-domain correlation for chirp is done at a sampling rate of 2MHz and the result is shown in Fig. 5.2. As mentioned in Sec. 2.4, 2MHz is providing with an accurate digital footprint of the received signal but in-order to conform with the hardware processing capability the locally generated signal is generated with lowered sampling rate to observe the effects for reduced sampling rate.

Single pronounced peak is observed until 500KHz but with further reduction of sample rate, multiple peaks are found. This could make the detection poorer, however if the maximum correlation value is considered for lower sampling rate, the error of detection from that of sample rate of 500KHz or higher would be a negligible $5\mu s$. The first reflected signal is comparable to the line-of-sight detection and the time difference between the two is roughly $30\mu s$.

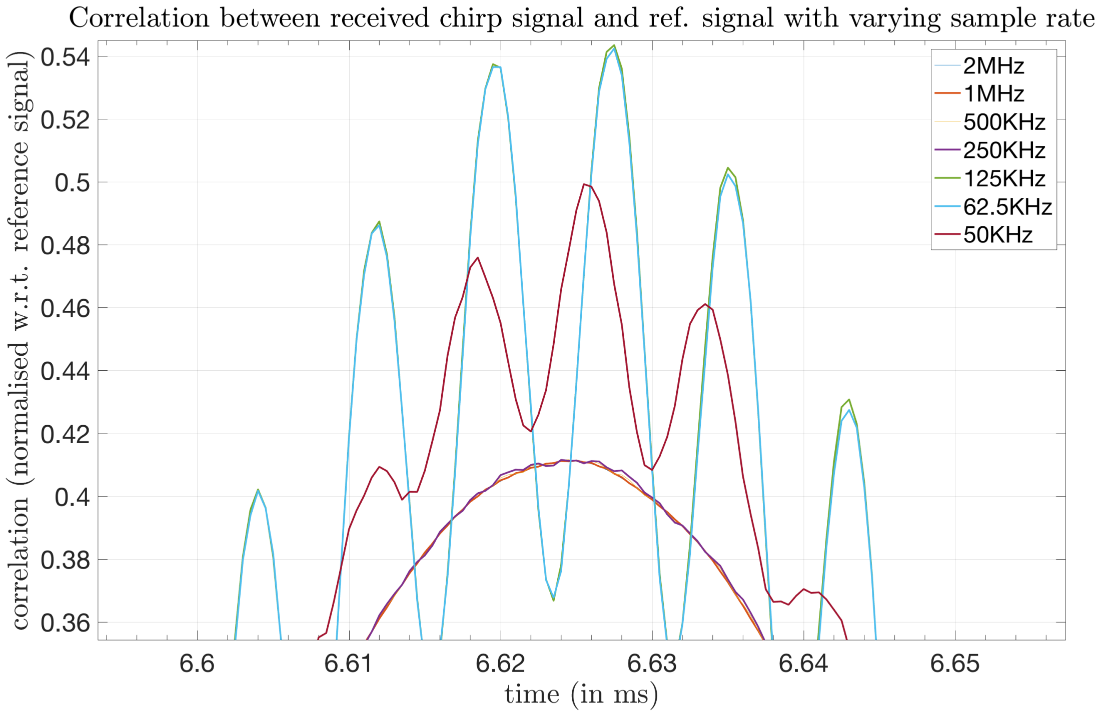
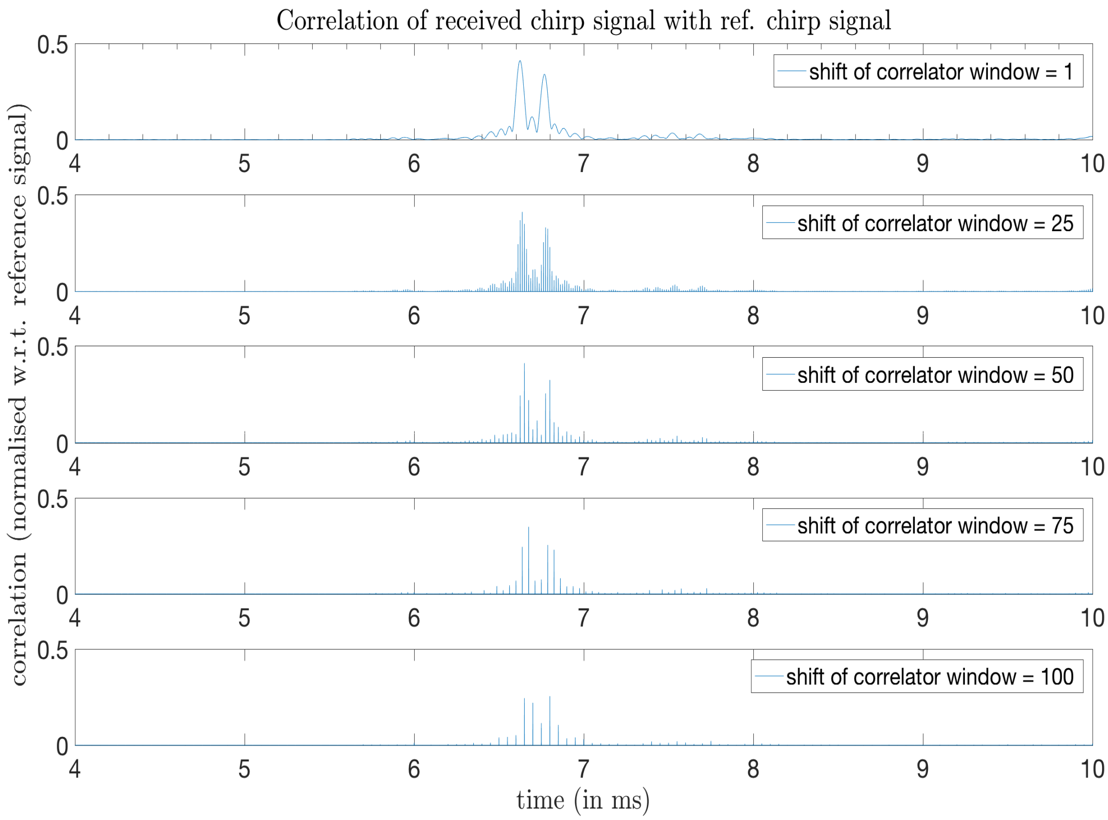


Figure 5.3: Variation of peak of correlation result with lowered sample-rate of the reference signal, when correlated with the received linear up-chirp signal

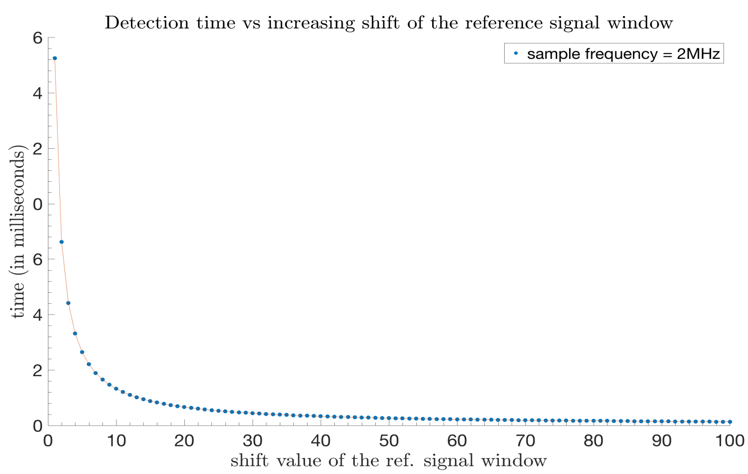
The correlation is also evaluated with varying shift the reference signal window to enhance the computation time. In experiment, the signal window is shifted with a step-size of 25μs as shown in Fig. 5.4 which conform with the signal threshold of 0.35, as mentioned earlier in this section, for optimum detection.

The implementation of step-size improved the signal detection time by 6.49ms when correlation using a step size of 50 (equivalent to a shift of 25μs) is compared with correlation performed with step-size 1. From Fig. 5.4b, it is observed that the detection time is saturated with the shift value increased from 50 onwards and moreover it has been seen that when the step size is increased more than 80 or the reference signal window is shifted more than 40μs, the peak of the correlation starts decreasing and the first reflection is getting stronger.

The delay in reception is another criterion on which the chirp and the FSK modulation were compared and a plot is shown in Fig. 5.5 with average delay in reception in each iteration. In case of chirp, an average delay of 3.3ms is found whereas in case of FSK, a delay of 0.32s is averaged



(a)



(b)

Figure 5.4: (a) Correlation between received chirp signal with reference chirp when the reference signal window is shifted by different value. **(b)** Variation in detection time of the LOS received linear chirp signal when correlated with a varying shift of reference signal window at the correlator keeping the sample frequency fixed at 2MHz.

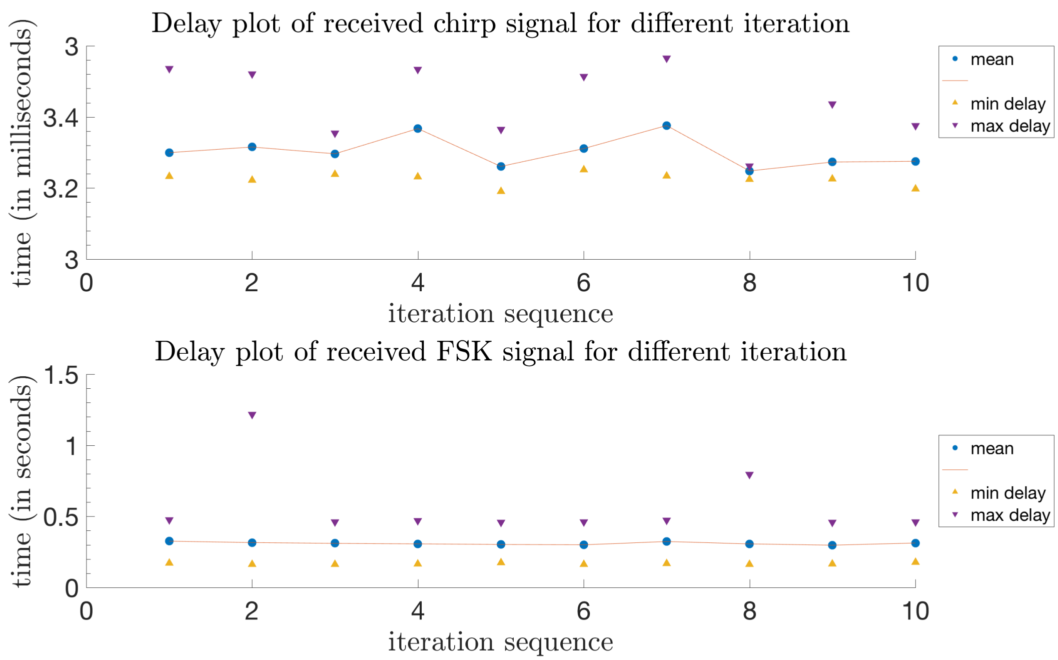


Figure 5.5: (Top to bottom) Average delay in each iteration obtained from received up-chirp along with maximum and minimum delay and bottom is the same for a received FSK signal. For chirp signal a bandwidth of 5KHz is considered and both the signal is of 1ms duration.

with a maximum delay of 1.2s is recorded in 2^{nd} iteration, proving that the detection of chirp signal is faster along with more accuracy rather than FSK counterpart.

The reason for such high delay in case of FSK received signal because it suffers in highly frequency selective channels and the rapidly varying surface interactions causing multi-path reflections. These is why a peak value is hard to find and implementation of a threshold detection becomes important. Hence a later reflection being dominant and that reflected peak being the strongest is chosen over the line of sight peak found from the correlator. This also gives rise to threshold ambiguity and it is already observed that the threshold detector can be defined as a function of post-amplifier gain.

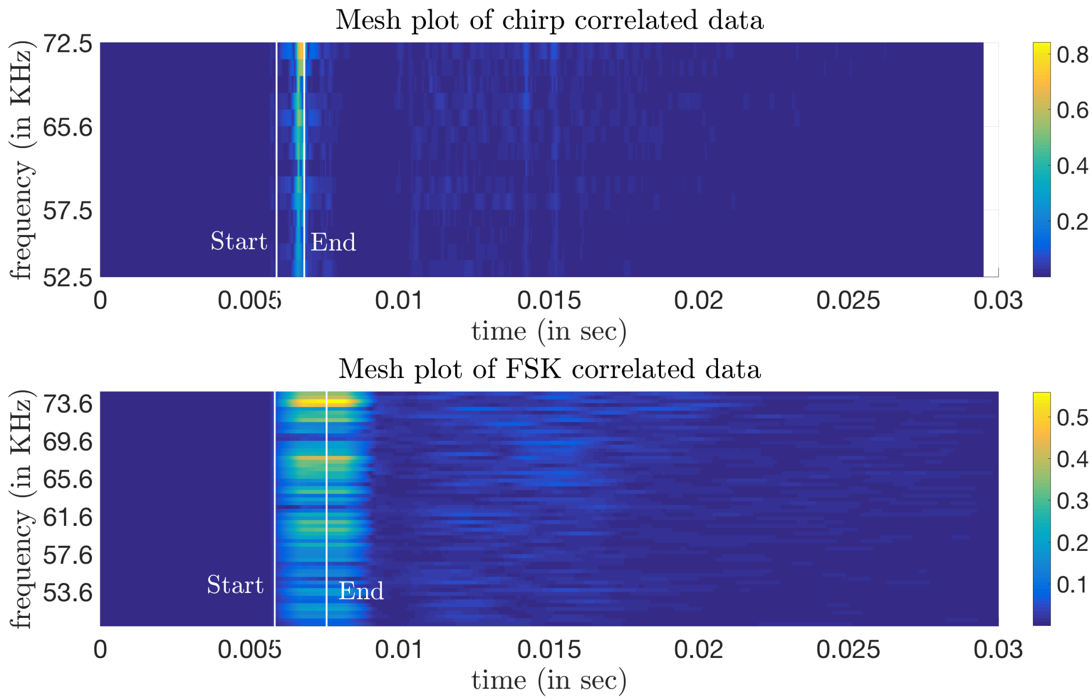


Figure 5.6: Received signal intensities for a symbol length of 1ms at a distance of 4m and a depth of 1m (correlation of received signal with reference data at receiver) when the experiment is conducted at channel Harburg.

The last part of this test involves the frequency component that is observed in the received signal and to compare the spread of the intensities as well as when the reception intensities is at a maximum. In Fig. 5.6, the frequencies intensities are observed for both FSK and chirp received signal. In case of chirp the maximum intensity is peaked at a single point of time marked with end and the multi-path effect is can be distinguished from the fundamental signal whereas in case of FSK the maximum intensity, also marked by end, is observed along with the multipath effect which marred the FSK modulation technique is visualized. The term end denotes the maximal value of the cross correlation which occurs at the point when whole line-of-sight signal is received.

5.2 Außenmühle Lake Test

After the test at channel Harburg test, the idea of Doppler shift was explored as the results were

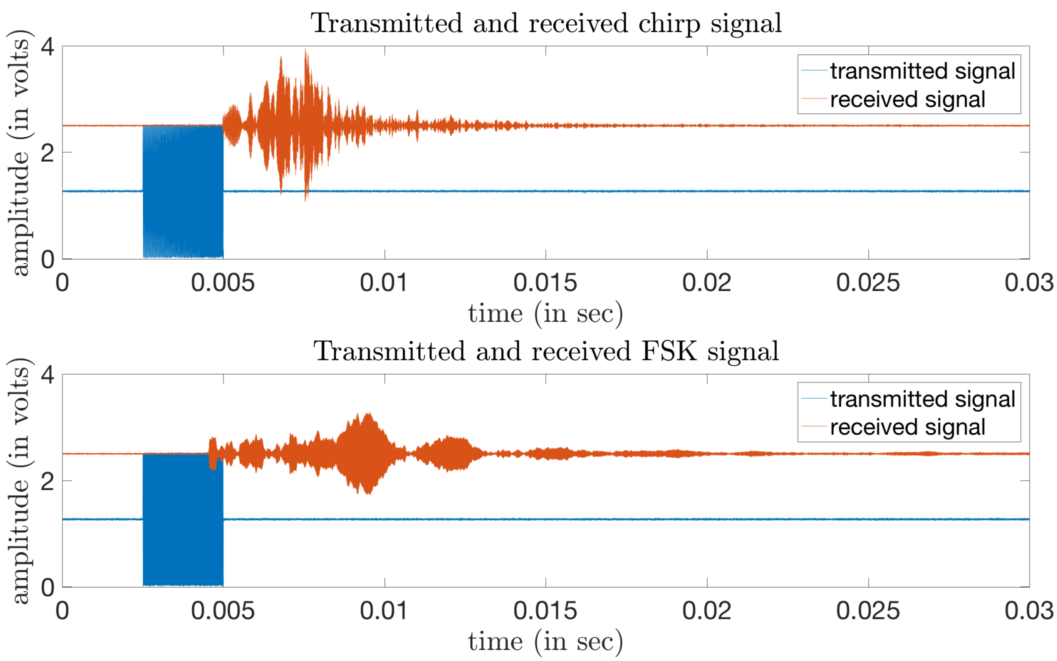


Figure 5.7: The chirp and FSK signal is created with 2.5ms signal duration and chirp signal has a bandwidth of 25KHz and a starting frequency of 50KHz. The FSK has a carrier frequency of 74.8KHz. The post-amplifier gain is set at 0dB.

found satisfactory in favor of chirp modulation than its counterpart FSK modulation. Sec. 2.5.3, briefed the idea of Doppler effect and is supported by [21] which further motivated the idea of using linear chirp for underwater acoustic communication and this being the main reason of testing linear chirp in Außenmühle lake in Harburg and compare the same with FSK to visualize the improvement as theoretically claimed.

The lake is chosen rather than the channel is merely due to a change in environment and the experiment was performed on a boardwalk and the transmitter hydrophone is kept fixed while the receiver hydrophone is moved at a velocity of 63 meter/hour or 0.017 m/s. The speed of sound is again taken as 1482m/s as per the Fig. 2.7. So, according to Sec. 2.5.3 and Eq. 12, the Doppler frequency shift in this scenario is 0.57Hz. The transmitter and receiver hydrophone initially placed at a distance of 3.58 meters for the experimental setup. The signal duration of 2.5ms is considered for this scenario as according to [1], the state-of-the-art technique used the same because with

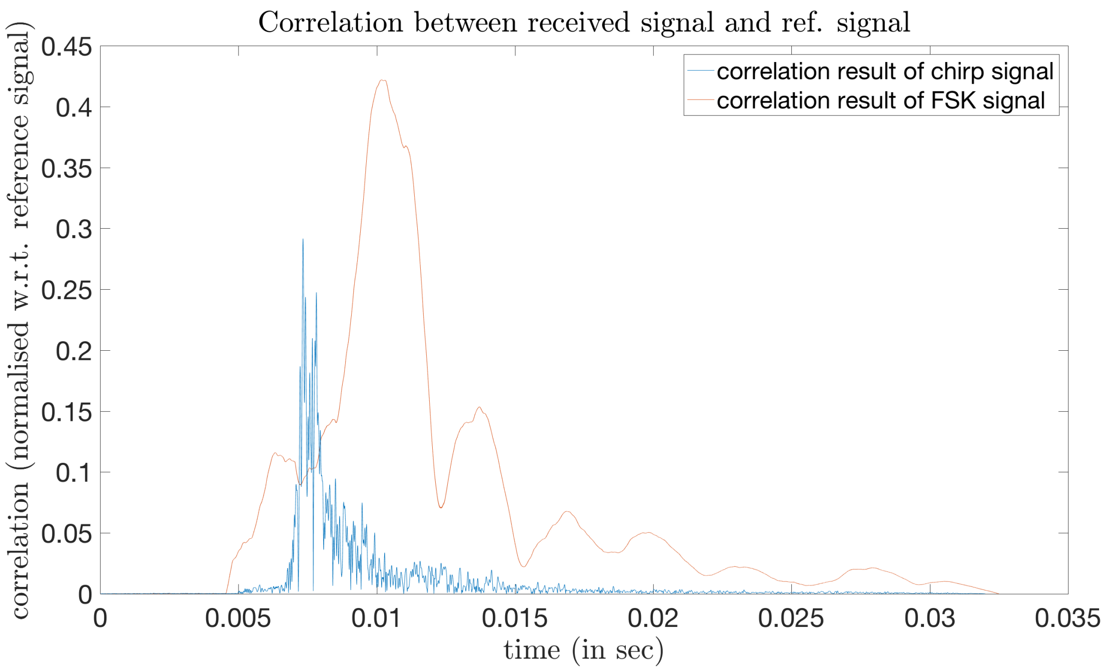


Figure 5.8: Correlation of the received signal shown in Fig. 5.8 with the locally created reference signal with a-priori information when the receiver is moving at a speed of 63meter/hour with respect to the transmitter.

this signal duration and a frequency spacing of 400Hz, orthogonality between two consecutive carrier frequency is obtained. In Fig. 5.7, the observer, in this case the receiver, is hearing a lower frequency in the beginning before the actual LOS signal arrival.

Fig 5.8 is a result of the time-domain analysis, i.e. correlation, performed on the received chirp and FSK signal shown in Fig. 5.7. As the frequency deviation found out to be 0.57Hz, the FSK carrier frequency observed at the receiver would be 74.80057Hz whereas in case of chirp, the time-bandwidth product will remain the same and as described in [21], the said product is well below 2000, which makes it robust to Doppler shift.

In Fig. 5.8, it is observed that a prominent peak is obtained for chirp which is found at 7.33ms, the time approximately at which the signal is received and considering a threshold value of o.25, the signal can be spontaneously detected. In case of FSK, because of the boundary interactions due to multi-path propagation, a delayed peak is found, which is also not prominent as compared to chirp.

Chapter 6

Evaluation

From the various tests performed in different environment, the primary observed point is a prominent peak when a chirp signal is considered for modulation of the transmitted bit. In the entirety of the project, signal duration of 1 ms is considered after initial consideration of 2 ms during the small-container test. The power spectral density is checked initially and 1ms symbol duration is decided upon for optimum use of limited usable bandwidth underwater. This is because the side-lobe suppression is better for 5 ms 25 KHz chirp signal duration although for 1 ms 25 KHz chirp signal it is comparable and is observed from the Fig. 4.4.

This is why, a further step is taken when a test is done in tank and the bandwidth used for each chirp signal is reduced to 5 KHz. Again, the PSD is consulted to check the side-lobe suppression is found to be even more narrowed down as the bandwidth is reduced and is seen from Fig. 4.9.

The step next taken when the experiment is performed outdoor in lake and channel in Harburg to compare chirp with FSK modulation and is primarily analyzed in time-domain on the basis of the time at which the signal is detected and the delay in detection. Moreover, the robustness of chirp is tested by moving the transmitter at a very low speed of 63m/h and is compared with FSK.

In case of the small-container, linear chirp with 5 ms duration and bandwidth 25 KHz with starting frequency 50 KHz worked best as it mitigated the severe reflected environment of the small container. This scenario suppressed the side-lobes to a greatest extent and hence the main lobe is most prominent among others. The argument is presented here is based on the increasing time-bandwidth product and in this particular scenario the time-bandwidth product achieves a maximum value of 125 among all considered scenarios in small-container test and is shown in Fig. 6.1. The correlation value exceeds 1 because it is normalized only with respect to the reference signal.

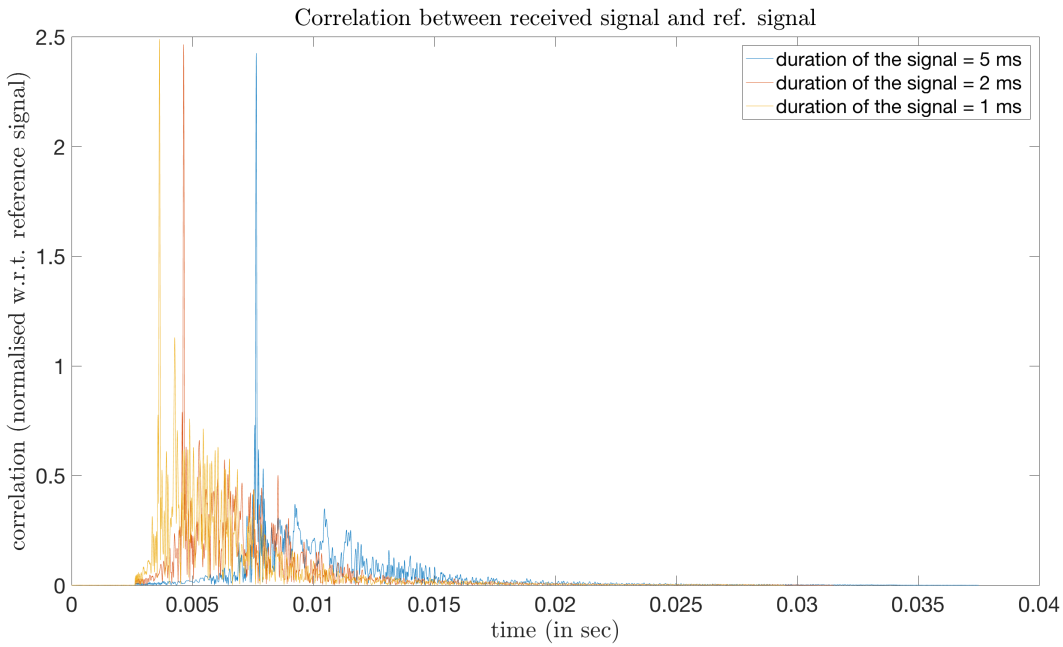


Figure 6.1: Correlation between the received signal and the locally stored signal with different signal duration and same bandwidth of 25 KHz with the same starting frequency of 50 KHz and the test is performed in the tank at Hamburg Institute of Technology with post-amplifier gain set at 16 dB.

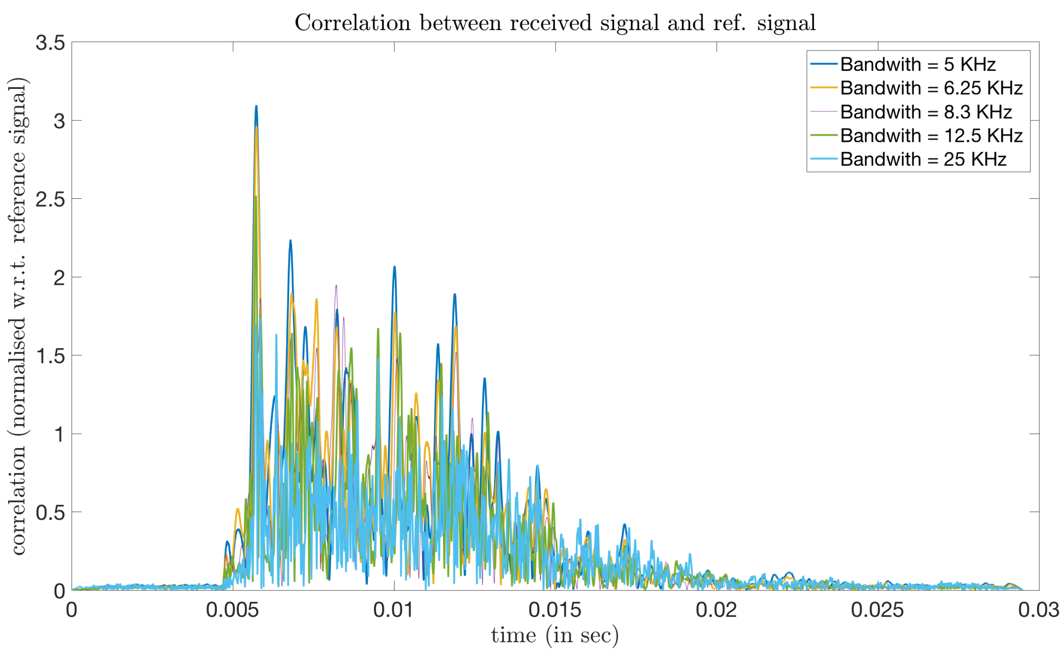


Figure 6.2: Correlation between the received signal and the locally stored signal with same signal duration but different bandwidth of 25 KHz with the same starting frequency of 50 KHz and the test is performed in the tank at Hamburg Institute of Technology with post-amplifier gain set at 16 dB.

Afterwards test is performed in the tank and the Fig 4.8 is depicted in the same plot to evaluate the scenario elaborately. The best scenario obtained in case of tank is 1 ms and with bandwidth with starting frequency 50 KHz. The strength of the first reflected echo signals compared to the line of sight peak and decreases as the bandwidth decreases keeping the signal duration same which guides to pinpoint this scenario as the best among the other considerations. The flipside of this scenario is that the narrow bandwidth might be exploited better and 1 ms is considered as to reduce the computation time at the receiver as the side lobe compression is 10 dB lesser compared to 5 ms chirp signal.

Later, with the elementary idea with the chirp behavior, the test in done in a channel with the same parameters as taken in case of tank test and additionally FSK is also tested for the exact setup to compare between the two modulation techniques and to select the best possible setup in case of open fresh water scenario as in channel Harburg. In this scenario, an interesting feature was observed contrasting the results obtained. The side-lobe suppression is again found comparable across every bandwidth when the chirp duration was fixed at 1 ms but from Figure 6.3, peak is found to be maximum for bandwidth 25 KHz and decreased as the bandwidth decreases. In this scenario, the 25 KHz linear chirp provides with the excellent peak for the LOS symbol however the chirp signal with 5 KHz may also provide comparable decision clarity with proper thresholding.

From Fig. 6.3 and 6.4, it is visualized that while a peak is observed for every bandwidth in case of chirp, a flat or smooth slope is observed for FSK reception when the intended transmitted signal is received completely. Moreover, due to the boundary interactions, the roof top of the correlation is not similar for every carrier frequency which further leads to the ambiguity on the decision of boundary detection.

Thus, arguably chirp performs better in open environment even with lowest time-bandwidth product of 5 amongst all scenarios under consideration.

Lastly, another environment is chosen for the Doppler shift test. The final test was performed in a lake in Harburg to have a different environment with weeds and higher level of particles which gave rise to scattering. Additionally, people were rowing which provides additional noise and water surface turbulence, and hence is considered to be close to a real-time situation. The channel Harburg was relatively calm as only a few people fishing when the test was performed and there were no boating or shipping activity taking place at that particular time.

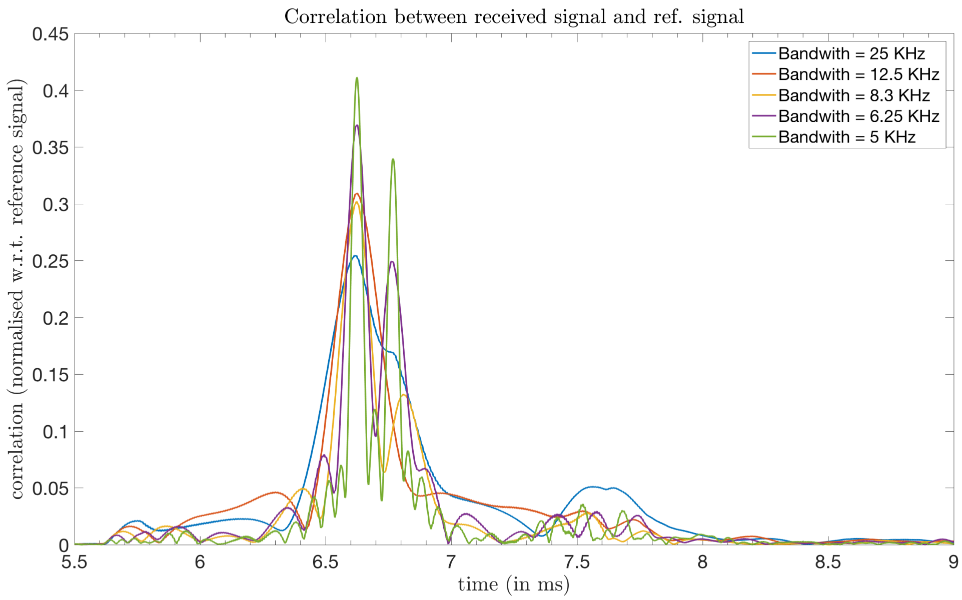


Figure 6.3: Correlation between the received signal and the locally stored chirp signal with same signal duration but different bandwidth of 25 KHz with the same starting frequency of 50 KHz and the test is performed in channel Harburg with post-amplifier gain set at 0 dB.

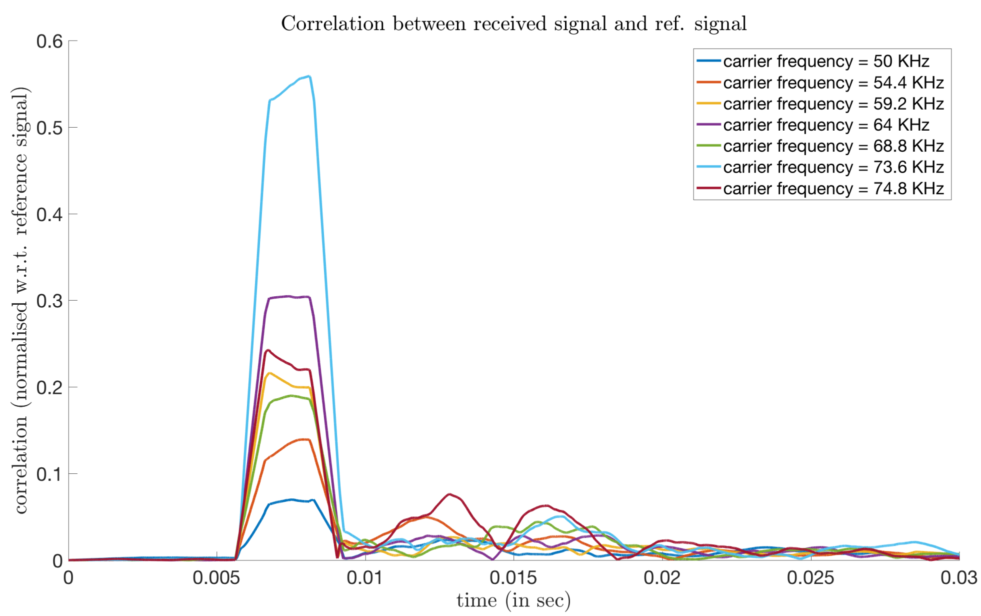


Figure 6.4: Correlation between the received signal and the locally stored FSK signal with same signal duration but different carrier frequency and the test is performed in channel Harburg with post-amplifier gain set at 0 dB.

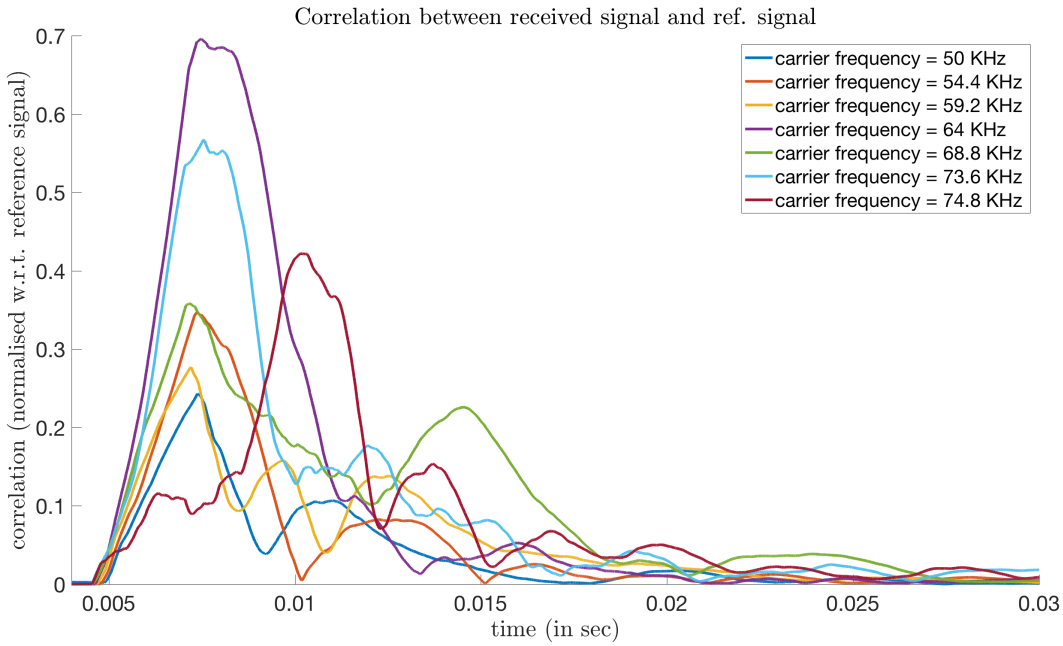


Figure 6.5: Correlation between the received signal and the locally stored FSK signal with same signal duration of 2.5 ms but different carrier frequency and the test is performed in Außenmühle lake, Harburg with the transmitter hydrophone moving at a speed of 63m/h with respect to the receiver hydrophone with post-amplifier gain set at 0 dB.

Doppler shift is already described in section 2.5.3 and the experimental setup is also elaborated in section 5.2. Again, the result of chirp modulation is visibly better even when the hydrophones' relative speed is as low as 63 m/h when comparing Fig. 6.5 with Fig. 6.6 and Fig. 6.7.

In case of FSK as shown In Fig. 6.5, the roof top of the correlation of FSK symbol is abrupt with no clear or prominent peak which is due to overlapping of closely placed echoes due to multipath propagation. Hence, it is impossible to find an exact time at which the signal is received when there is a movement of the hydrophones

In case of chirp, decreasing bandwidth makes the detection worse when the signal duration is kept constant at 1 ms which is shown is Fig. 6.6 and so it can safely be denoted that increasing time-bandwidth product effectively improves detection of the received signal and consequently the boundary detection of the preamble is easier.

Thereafter the impact of different symbol duration for the highest possible bandwidth is investigated and is shown in Fig. 6.7. From the correlation of chirp signal with a pulse duration of

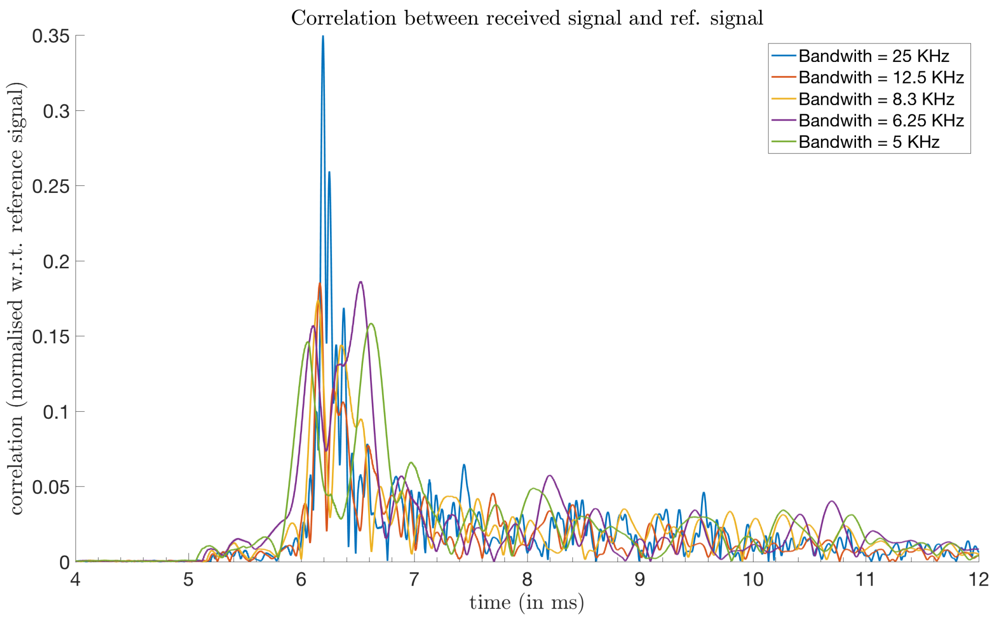


Figure 6.6: Correlation between the received signal and the locally stored chirp signal with same signal duration of 1ms but different bandwidth with same starting frequency 50 KHz and the test is performed in Außenmühle lake, Harburg with the transmitter moving at a speed of 63m/h with respect to the receiver hydrophone with post-amplifier gain set at 0 dB.

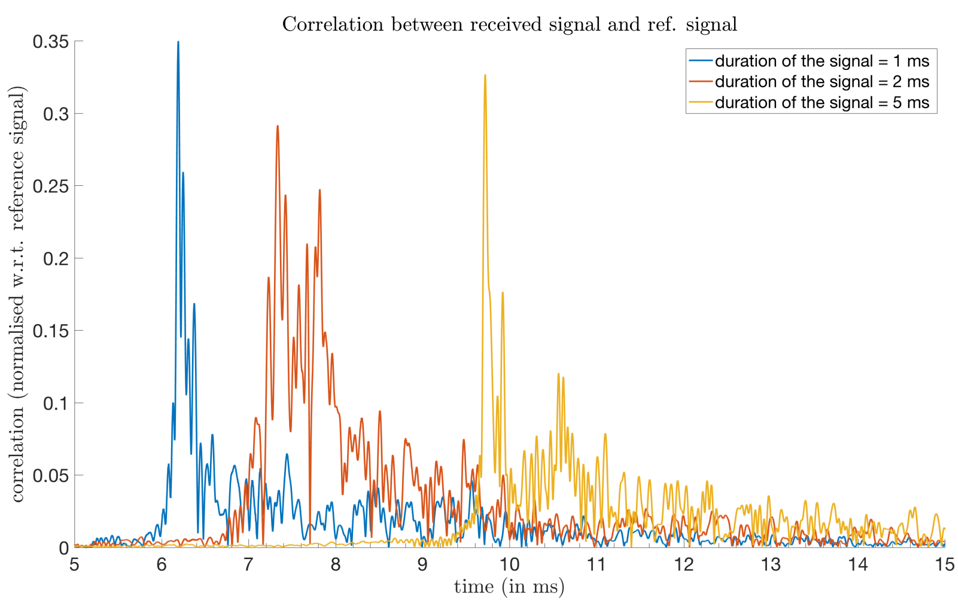


Figure 6.7: Correlation between the received signal and the locally stored chirp signal with different signal duration but same bandwidth of 25 KHz with same starting frequency of 50 KHz and the test is performed in Außenmühle lake, Harburg with the transmitter moving at a speed of 63m/h with respect to the receiver hydrophone with post-amplifier gain set at 0 dB.

2.5 ms is observed to be marred with reflections whereas chirp signals with pulse duration 1 ms and 5 ms seemed promising and subsequently the peak of both said pulse duration is arguably stronger than the strongest reflected signal. In case of the chirp pulse with duration 5 ms, the ratio of the peak to its strongest echo is greater than its 1 ms counterpart.

For the scenario with Doppler shift test, chirp is clearly edging over FSK with computationally efficient being the 1 ms chirp pulse with 25 KHz bandwidth and starting frequency of 50 KHz.

At this juncture, the efficient choice of thresholding is to be denoted to make the detection faster and unambiguous. It is already visualized that the correlation ratio is varying as the gain value of the post-amplifier at the receiver. This problem can be solved if this quantity is known to the receiver as an a-priori information. In the test cases performed in this experiment, if the post-amplifier is set at 0 dB, a threshold of 0.25 at the correlator will help in accurately detecting the received signal and consequently detection of the preamble boundary will be exact.

Chapter 7

Conclusion

From the research project, it can be said with confidence that chirp will operate better than FSK and offer a resilient communication technique in underwater acoustic communication. The reasons can be cited as that in case of the former method, a prominent peak is found rather than a triangular shaped region when correlation is performed and hence detection of time may be made accurately enough. Also, the delay in detection is much lower in chirp when compared to FSK which is an added advantage and it is also seen that chirp is multipath resistant and the detection delay at maximum found be a few milliseconds compared to seconds when FSK is considered over 10 iterations. As μAUVs are in constant motion underwater, the Doppler shift plays a very important role and the detection time of chirp received signal are found to be lower when compared to the FSK detection with same symbol duration.

The results showed that chirp will be better than FSK modulation when it comes to accuracy in detection and the ambiguity due to echo signals which is found when FSK is signals can be negotiated by applying chirp. The most important aspect that is confirmed is the robustness of chirp signals against multipath propagation and Doppler shift.

The time-bandwidth product is varied and its directly proportionality with the power concentration in the main lobe is also verified and helped in finding the optimum set of parameters in each case. The further work will comprise of implementation of the same in cortex M4 and to further optimize the parameters with respect to hardware limitations. This is because the signals are generated with a sampling rate of 2MHz and the sampling rate is reduced to 500KHz with an accurate detection but further reduction generates multiple peaks surround the point of actual detection which is still effective provide that the maximum correlation is identified but clear thresholding will not be possible. It is also intended to research how chirp modulation will behave if the relative velocity between the transmitter and receiver is in km/h range as during the experimental process due to limited resource the effect of Doppler shift is observed for a speed of 63 m/h.

Chapter 8

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Appendix A

Content of the DVD

The DVD contains the pdf version of the report and the source codes used for the project.